

HomelessAnalysis

AlejandroCespedesFernandez

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Introduction

For this analysis, we are going to be using the following data from the DHS collected from homeless shelters in NYC from July 2015 to March 2025. At the time of writing this article, this dataset was last updated on August 3rd, 2025. source: <https://catalog.data.gov/dataset/dhs-data-dashboard>

This data set is about the total amount of homeless people held in shelters at the time of the census. This dataset breaks down the population by age, ethnicity, and gender. For the final columns, there are the “shelters placed”

There are a lot of features in this data set, here is a deconstruction of the different variable names:

- FWC -> Families with Children
- AF -> Adult Families
- SA -> Single Adults
- HoH -> Head of Households

###Objective: The objective for this article breakdown is to analyze what census information leads to a housing placement from a temporary shelter. Each section goes into different census categories stated about, FwC, AF, and SA. The statistical model chosen will be a Poisson regression, as we will be working with a rate of placement.

NOTE TO READER

This data set changed halfway into 2022 its reporting frequency to be more frequent. To better explain the data, I decided to only keep the tail end of the data collected that month. This analysis is quite lengthy, so at the end I will be placing a results page where I discuss the findings all together and how they compare, and the conclusion section which will be similar to results but with more brevity. I hope you learn as much as I did during this paper, it has taken me quite a while to get everything together!

```
paste("The amount of data points we have for this data set is: ", length(data[,1]))
```

```
## [1] "The amount of data points we have for this data set is: 121"
```

Data Exploration

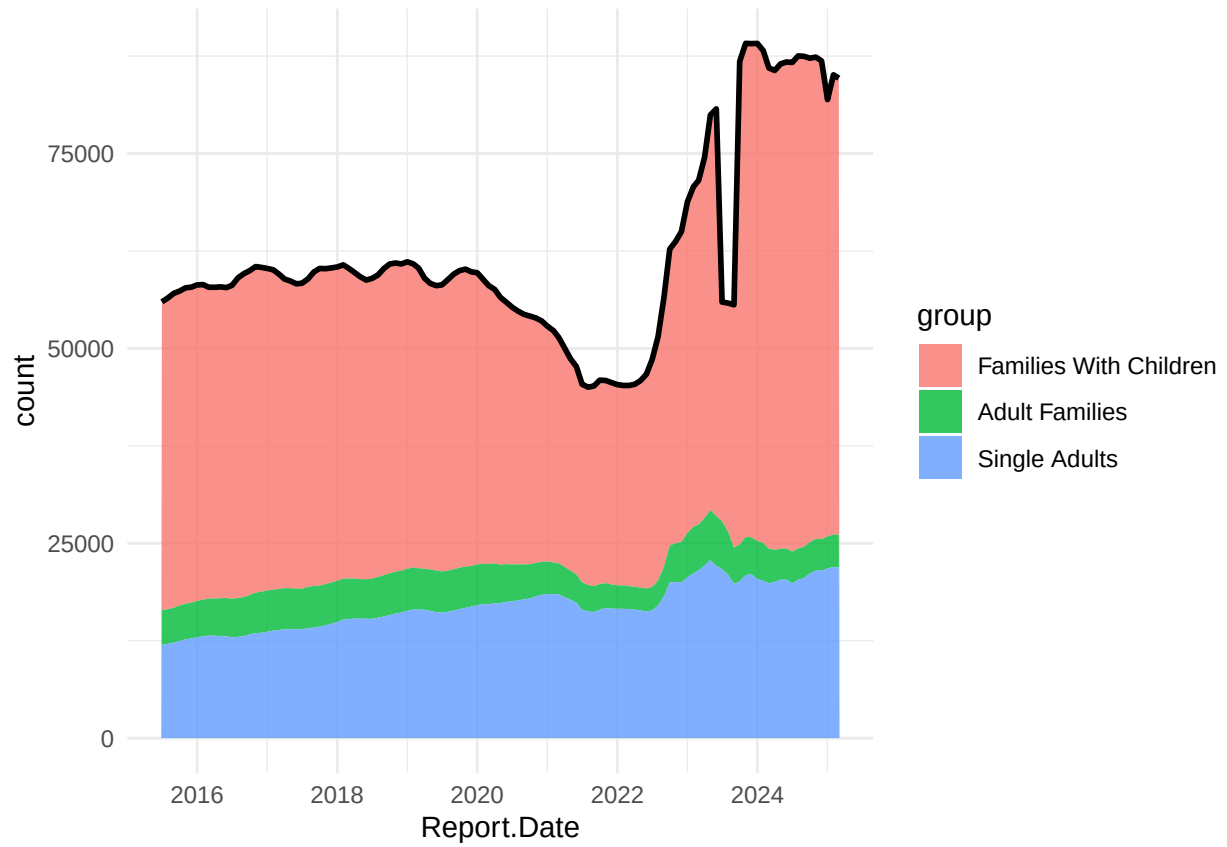
This data set mainly separates groups of people by age and race by the aforementioned categories: FWC,AF, and SA. There is a special category for school children, which can really only be applied for families with children.

Finally, we have some total shelter placements for the aforementioned groups.

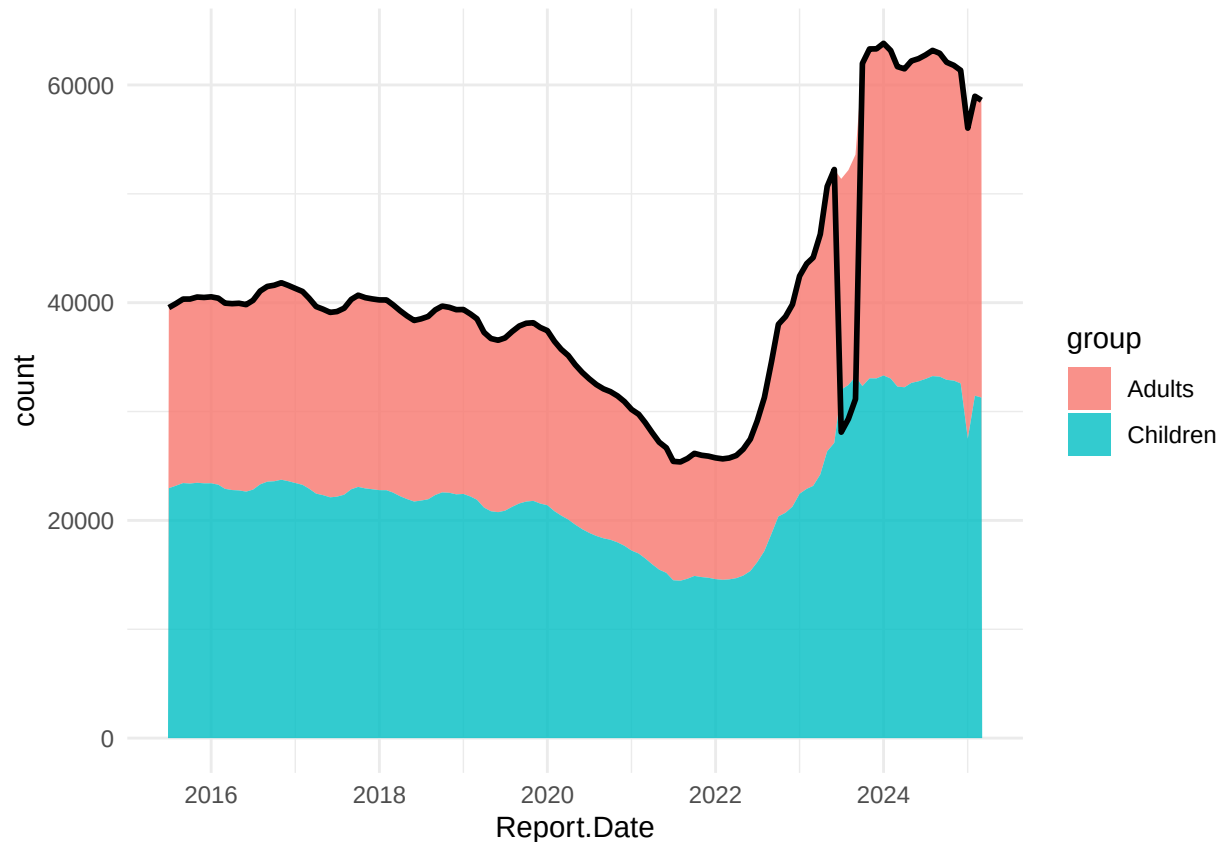
Breakdown by Age

Population Breakdown

Below is a graph that represents proportions per category of group.



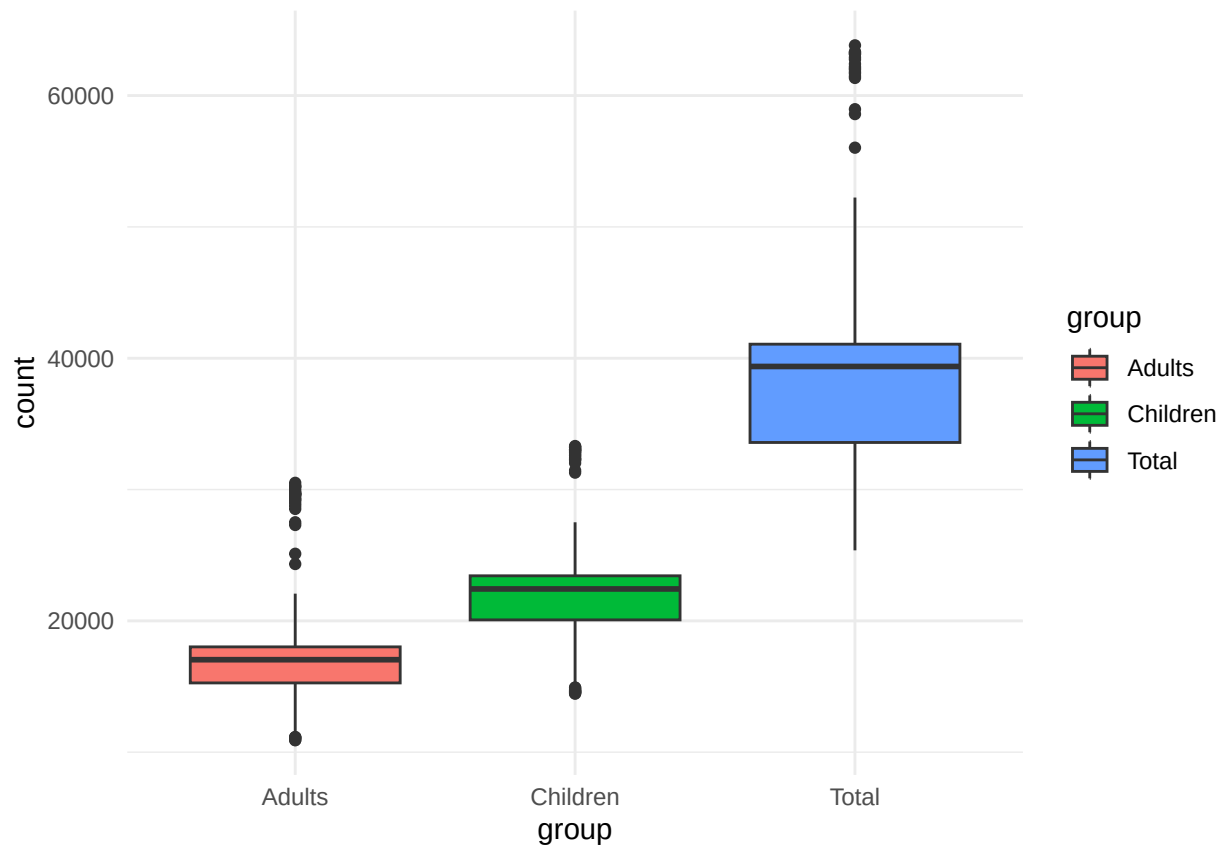
Families with Children by category (Adult, and Children)



There are a couple of interesting movements in this graph. First, it is obviously not a stationary graph. Second, note that the scale of this graph does not start at 0, so the scale of this graph is less dramatic as depicted.

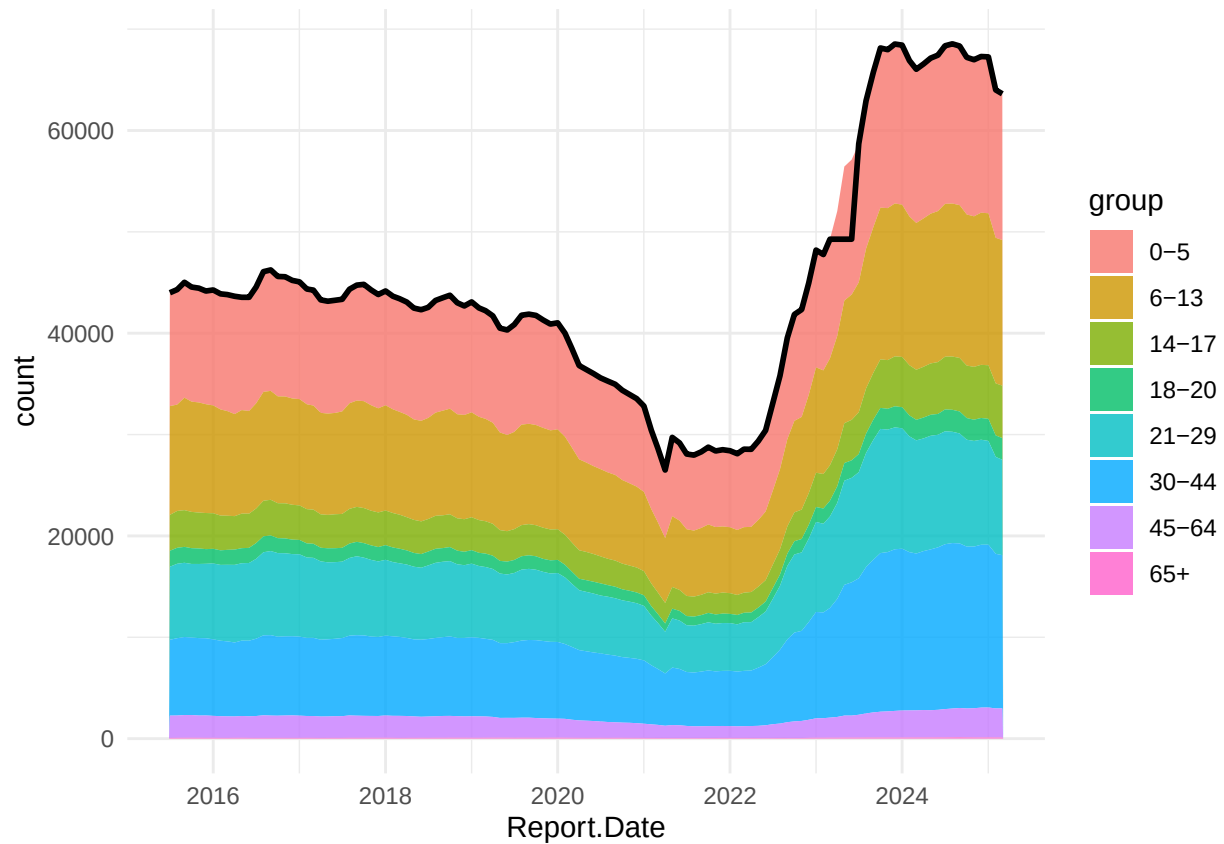
The first thing that jumps out of you on this graph is the decrease in homelessness in 2020, followed by an increase. Homelessness going down in 2020 makes sense, as that is when the 2020 pandemic happened. This unfortunately is probably correlated with [the increased rate of homelessness](#) that was seen in New York. The hyperlink discusses the mortality rate of covid-19 around April-May 2020, which lines up around the time of the graph. The numbers do not quite match up though, and the death of homeless persons was 22 (READ THE PAPER TO ELABORATE ON THIS)

Another feature of this graph is the appearance of a type of seasonality that appears before the covid-19 pandemic. Unfortunately, this is linked with the fact that [families face an increased cost of caring for their children](#), along with other costs such as increased electrical bills and the loss of support from when children are attending school. The seasonality argument is supported by the graph, as we see the seasonal affects of the main graph affect the children population a lot more than the adult population.



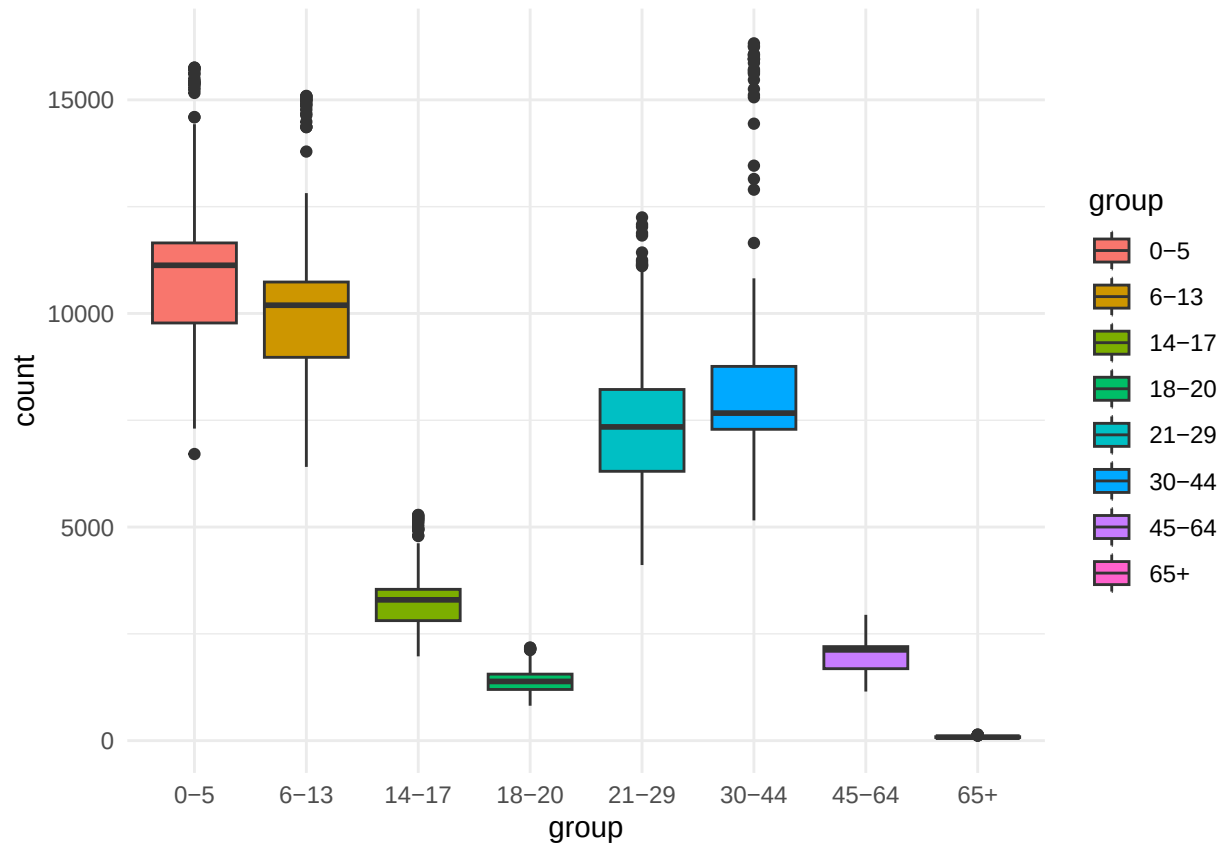
An alternate view of the stacked graph above. We can see that in families, children are more likely to end up homeless than adults. However, children are being more represented in this data just because families tend to have more than one child, and usually, the core family includes at most two people, the mother and the father.

Families with Children by Age



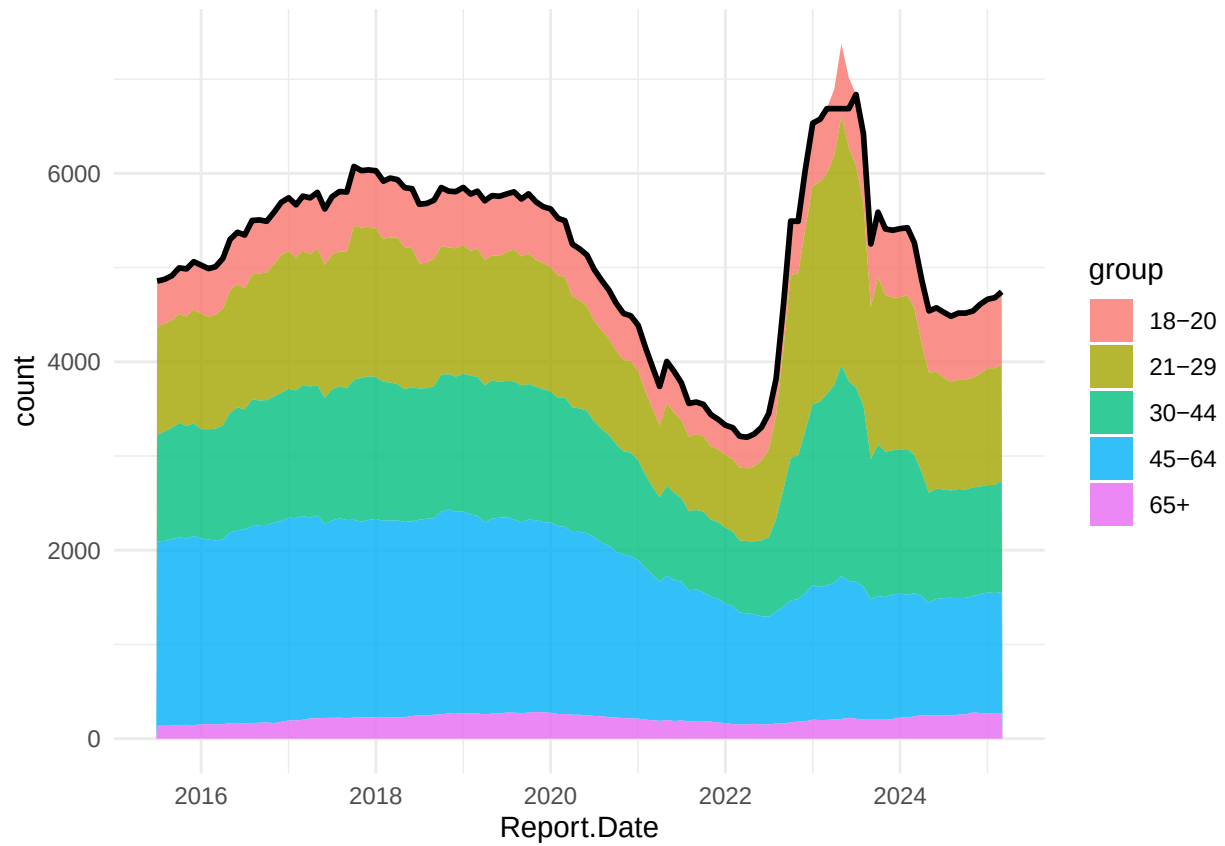
Breaking down the Families with Children demographics, we can see a breakdown of the majorities. Children aged 0-13 make up a huge chunk of homeless families. Reporting fell after 2020 with COVID, but soon shot up right after. This rapid increase in the homeless population can be attributed to the [influx of asylum seekers that have come to the USA](#). According to the source, that only accounts for 60% percent of the rise. The other 40%, remains unexplained.

However, if I were to interject, housing prices [inflated at a whopping ~7% in 2022](#). We can make a reasonable guess that since wages didn't keep up with this inflation, unfortunately a lot of people were not able to afford their homes.

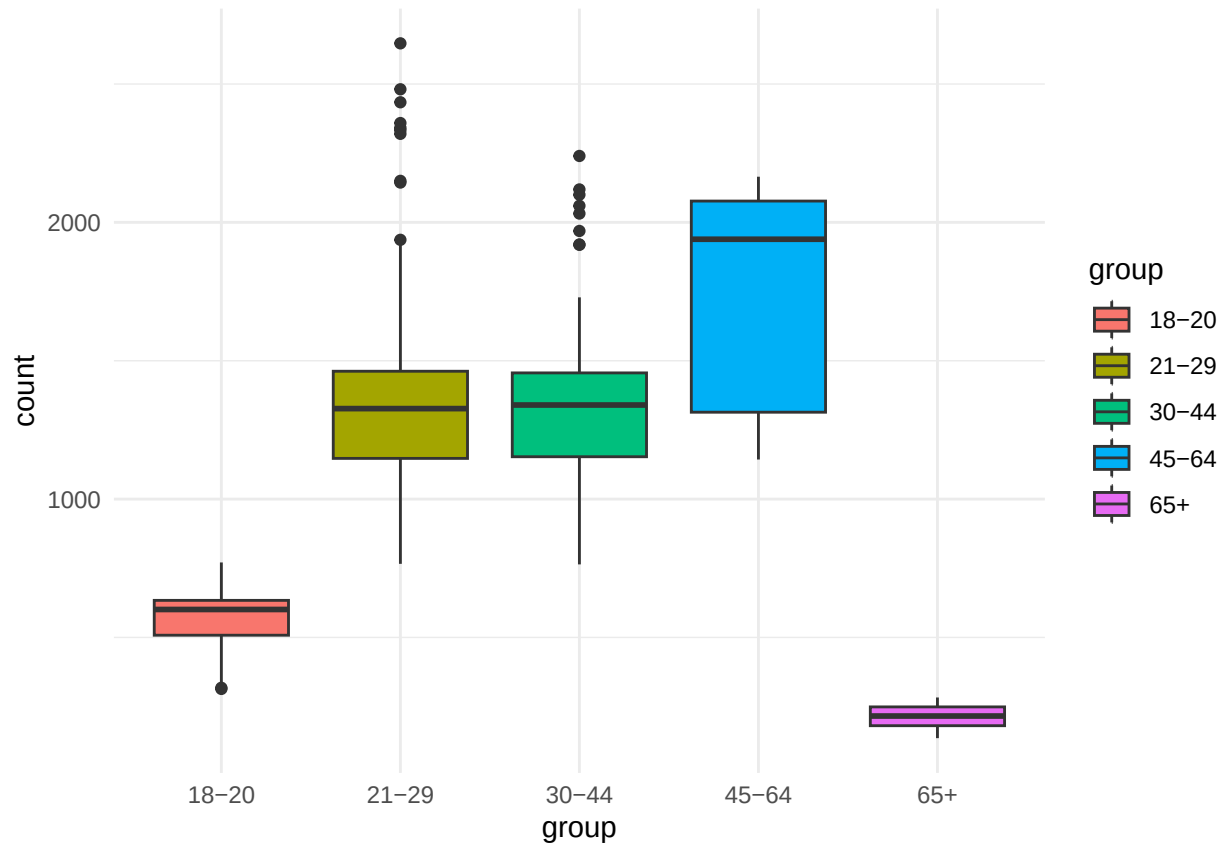


When we break it down by age, we can see that children make up a big part of the population when it comes to families with children. This is obvious, of course, but something we see is the second most prevalent age range, which is 21-29 and 30-44. Here, we can assume that these are the age of the parents. Then we see drop offs with 14-17, 18-20. Only assuming here, but based on the previous citation of needing assistance during the school year, it is likely at these ages, these groups are either helping the family with working on the side, or do not need as much financial assistance.

Adult Families by Age

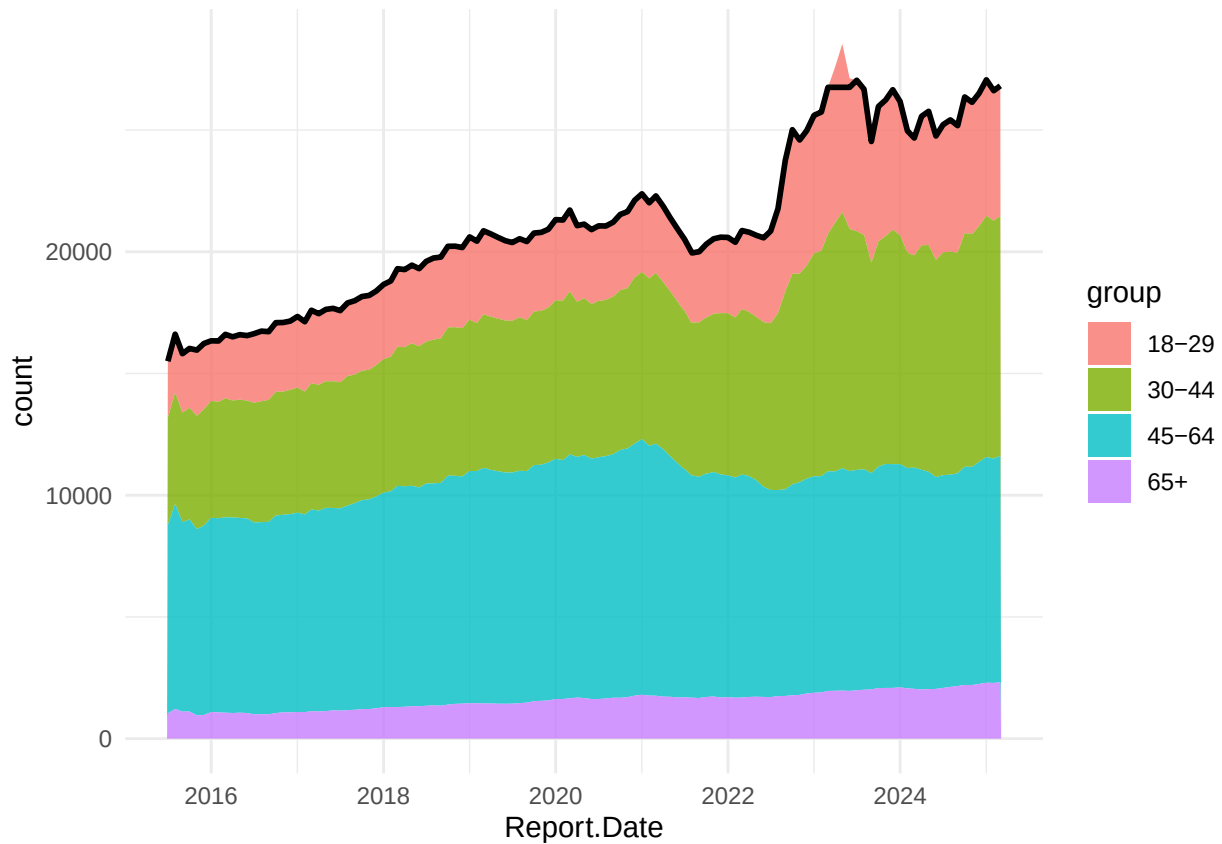


This graph shows the age range of Adult Families. We can see that a big chunk of the graph is made up of older generations, namely 45-64, and a follow-up of 30-44. We see a huge spike after 2022. This spike was also because of the spike in housing prices.



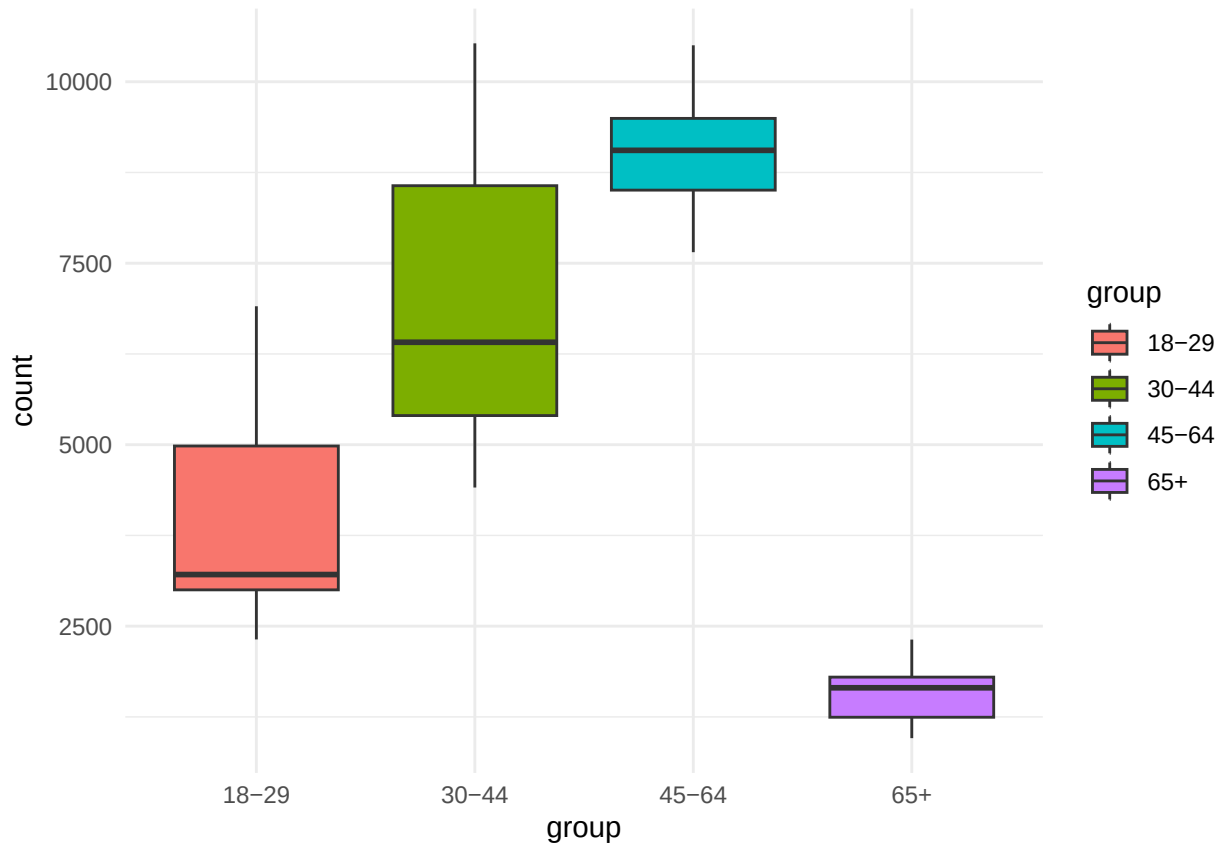
Here, the graphs reflect the volumes discussed with the previous graph. Adult families usually consist of caretakers taking care of dependents. From this graph, the majority of the concentration from the homeless population is 21-64, leaning towards 45-64. This is most likely because this is when people are independent, and have the least supports available to them.

Single Adults by Age



interesting notes: This didn't go down as far as the family graphs. In fact, this graph reinforces the central thesis of the article that families were more likely to get help during the pandemic (citation needed!), while single adults were less likely to get help.

The graph for single adults has an interesting change compared to the other graphs, the displacement during covid was not as severe. Some theories that could potentially affect this less aggressive reduction is that single adults were less likely to receive financial aid during COVID, so their living situation was unlikely to have changed as much. This also means that when COVID assistance ended, single adults were not affected as much. It is also the fact that single adults are more likely to find housing, and can make do with smaller or less desirable accommodations.



When we see the breakdown by age, what is interesting is that a really big chunk of the population has steadily been the group 45-64. while the age groups 18-29 and 30-44 have more variance between them. In other words, younger groups are more likely to enter and leave homelessness, while those 45-64 are more likely to be permanently homeless, or at least for a much longer time.

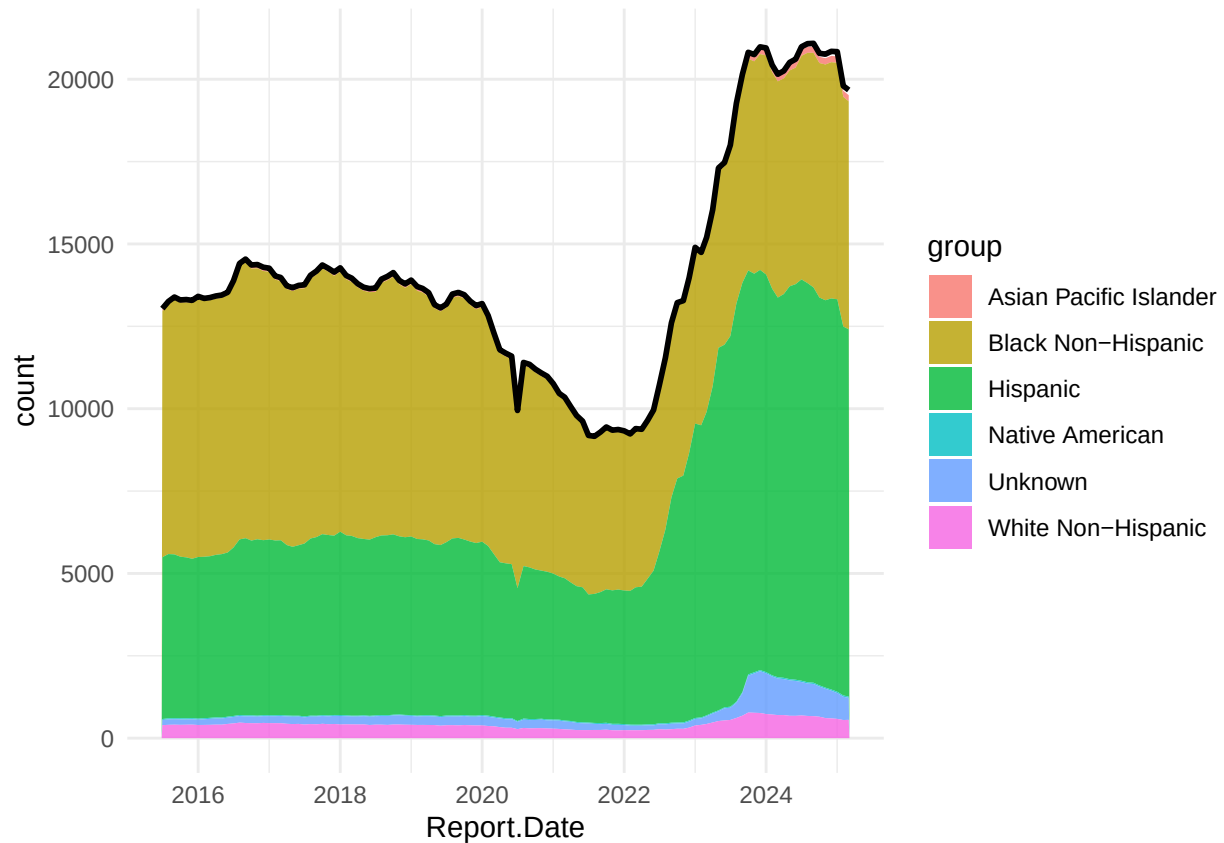
According to [the journal of The National Library of Medicine](#), this age group is the most vulnerable because of medical conditions that are harder to recover from. From the article,

Older adults with earlier experiences of homelessness have more adverse childhood experiences, chronic m

In other words, people who start off disadvantaged unfortunately stay disadvantaged. From that same article, it is also reported that payments from the government, namely SSI and SSDI payments, are not enough to pay rent. So after being overburdened, they are often left with no recourse and end up homeless.

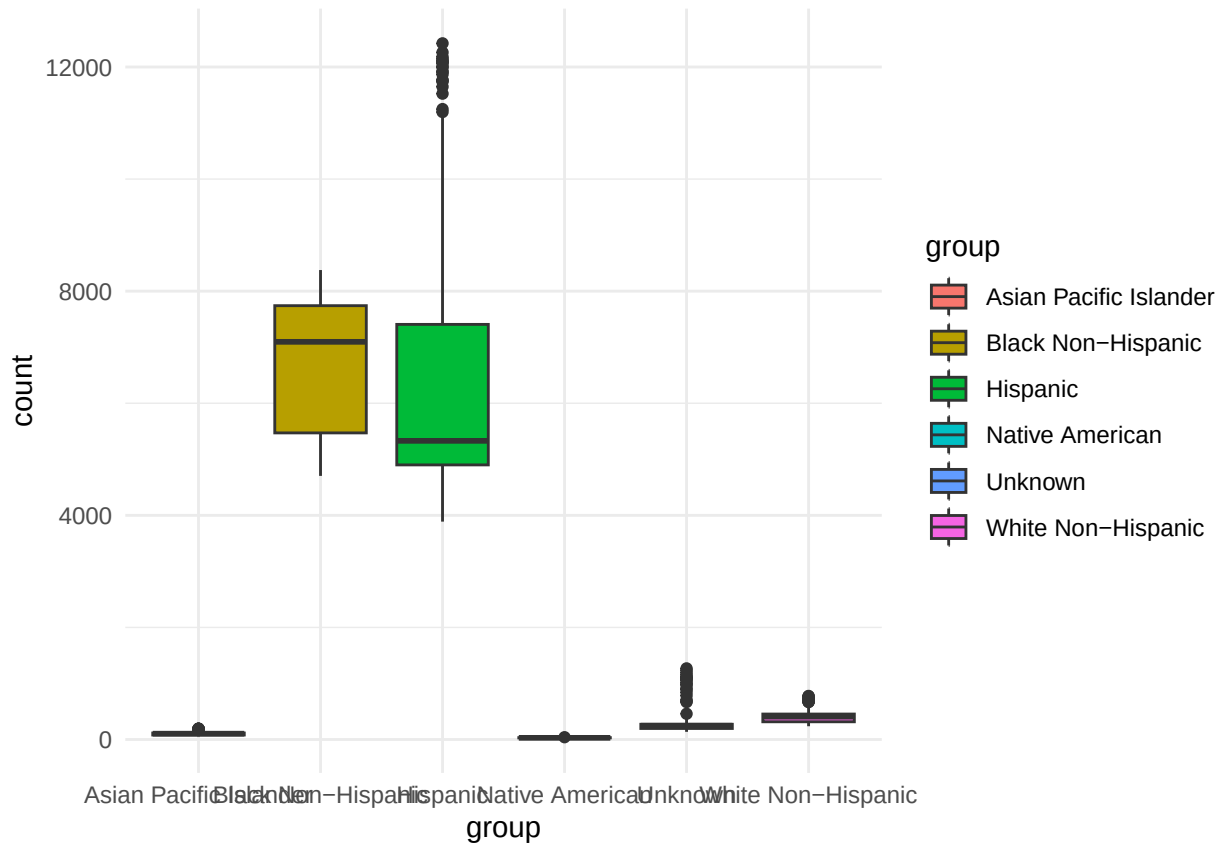
Breakdown by Ethnicity

In the section before, we were breaking the data down by age. The dataset also provides a breakdown of homelessness statistics by ethnicity.



This graph paints a clear picture towards structural inequality, with a shock in 2022. I would like to discuss the shock in the Hispanic population first. This shock first comes from the migration surge that New York City experienced in 2022, leading to many immigrants not being able to find housing in the city. If anyone remembers during that time, New York City had to set up [tents to accomodate all of the new migrants](#). The sudden shock from Hispanic migrants seems to be slightly decreasing, but if we were to apply a time series regression to this model, we can most likely conclude this shock is permanent.

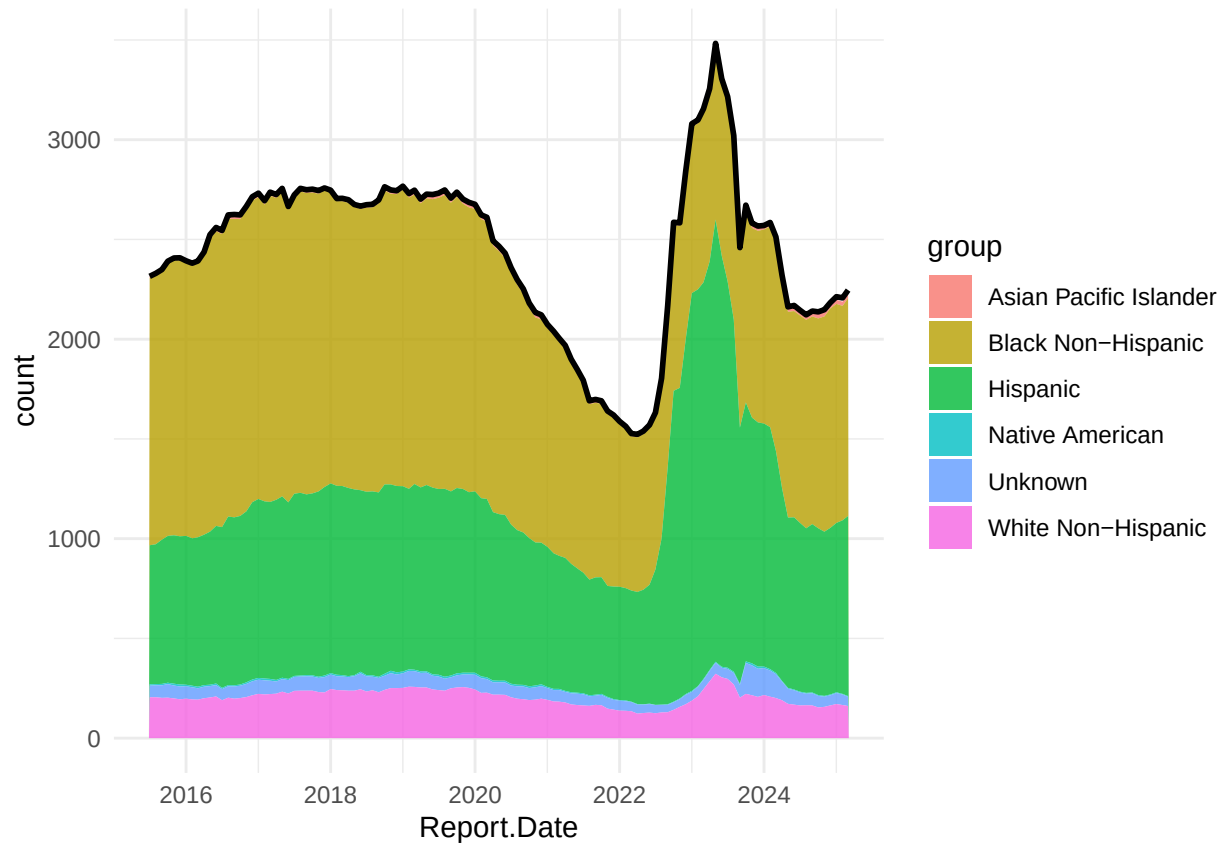
Some other things to note how low the white population is compared to the other ethnicity, it does not seem to match the proportions of New York City at all. This is a strong indicator of some kind of skew, namely structural inequality. However, I feel the following boxplot portrays this much better, so I will go into detail there.



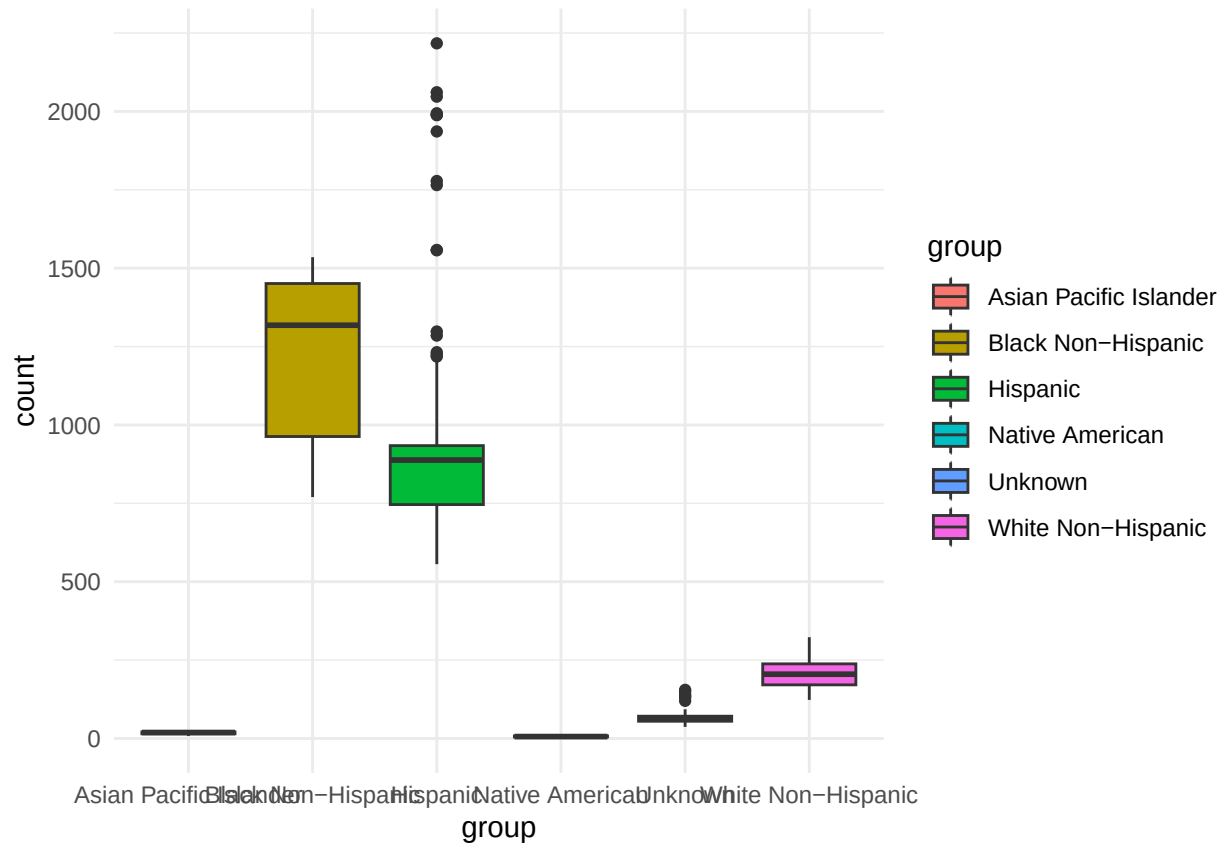
Here we can see that there is a clear disparity between the different ethnicities of New York City. [In this paper from Molly K. Richard](#), structural racism is the culprit behind these huge disparities. I don't want to dive into this topic too much in this paper, as this is beyond my scope, but basically structural racism is defined as leftovers left from racist periods in a country's history that still leave lasting effects to this day.

This paper goes into a couple of reasons. The biggest structural challenge towards Black and Hispanic populations is rent. More specifically, it seems there might be a relationship between homelessness and low vacancy rates. In other words, how available is housing? Recently, New York City has been building lots of luxury apartments, but barely anything for middle and lower class residents. What is completely outrageous is that many of these apartments are not even being used as residencies, [but as an investment](#).

Moreover, in the paper it is stated that Black and Hispanic people tend to face discrimination when searching for housing, which in turn keeps reinforcing the negative feedback loop. This feedback loop keeps Black and Hispanic people in poverty. Another feedback loop is criminalization. As stated before when discussing age brackets, those who have been incarcerated, especially at younger ages, tend to have fewer opportunities to leave that hole.



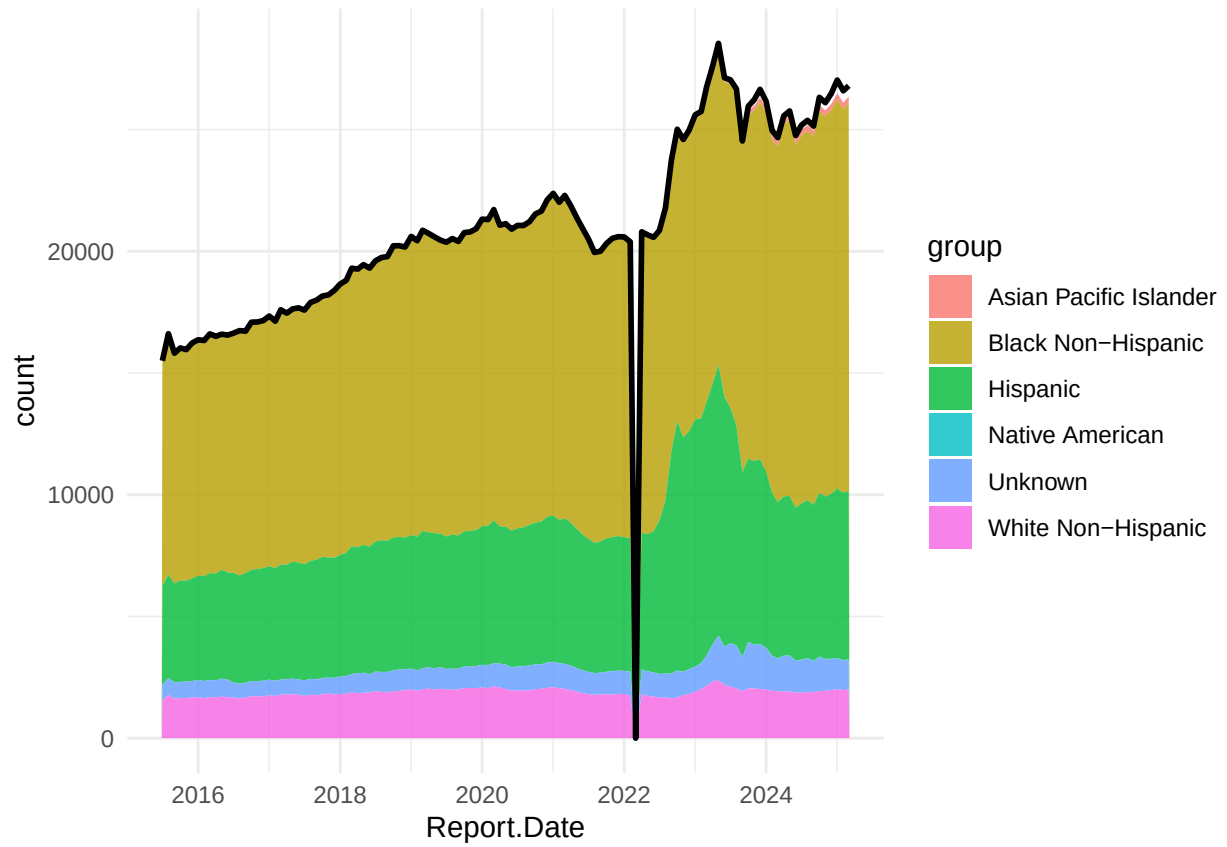
The same pattern appears with adult families, not much more to say here, other than the same spoke appears with the spike in immigration New York (and by extension, the USA) experienced.



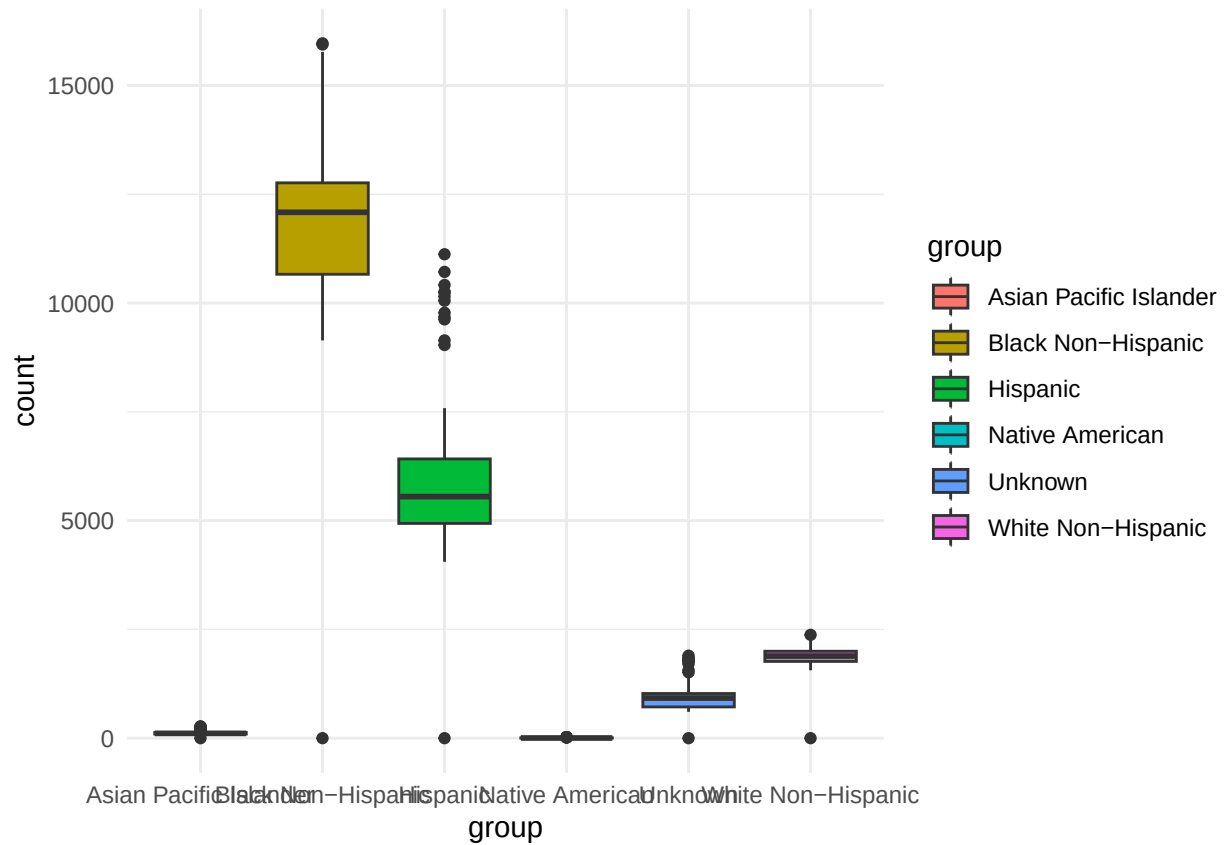
Same pattern as with Adult families, although some key differences. We see the same heavy tail with the Hispanic population. This is most likely from the hispanic spike in immigration.

Unlike the previous boxplot, here we see an increase in homelessness with white non-hispanic population.

```
## Warning: Removed 1 row containing non-finite outside the scale range
## ('stat_align()').
```

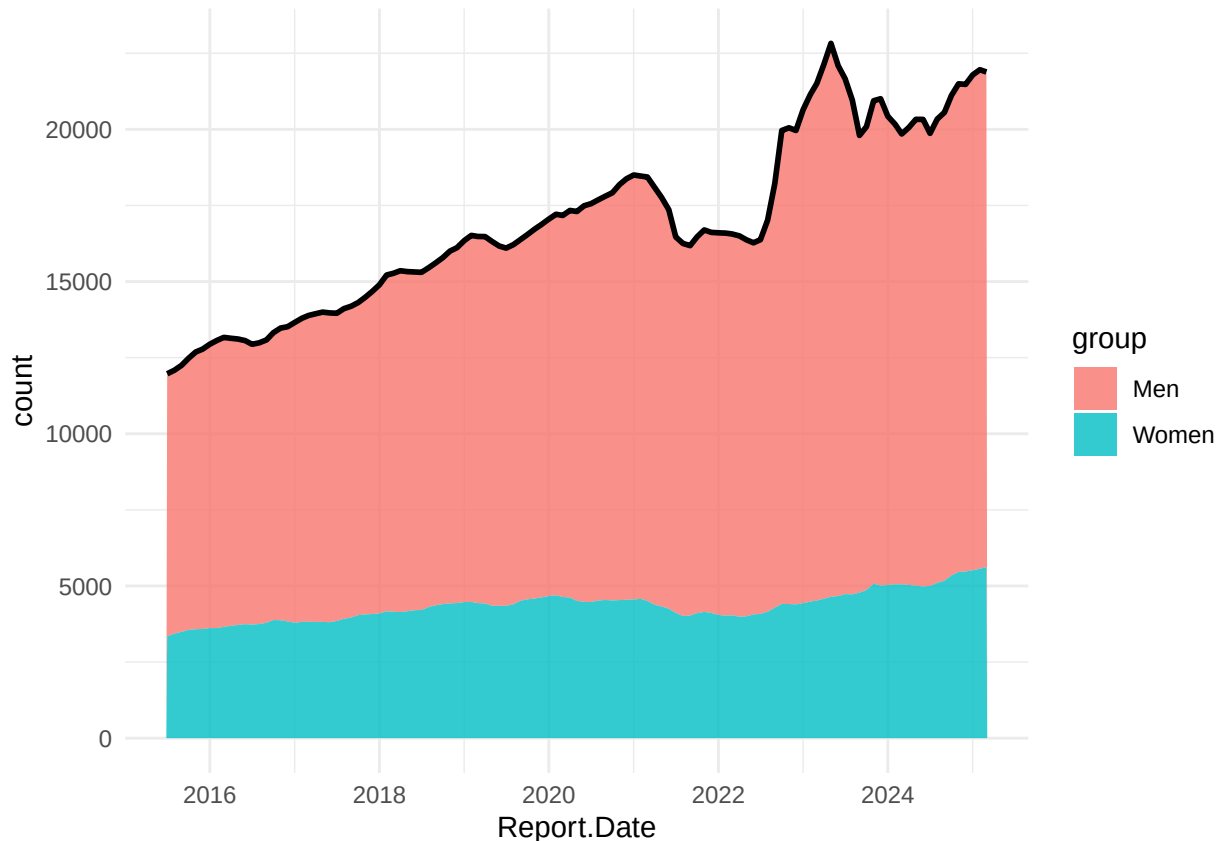


```
## Warning: Removed 1 row containing non-finite outside the scale range
## ('stat_boxplot()').
```



Now this is different from the other box plots, the black non-hispanic group are way more likely to be homeless. We saw the same pattern somewhat emerging with the adult families box plot, but here it is a lot more pronounced.

(why?) o ### Gender Breakdown



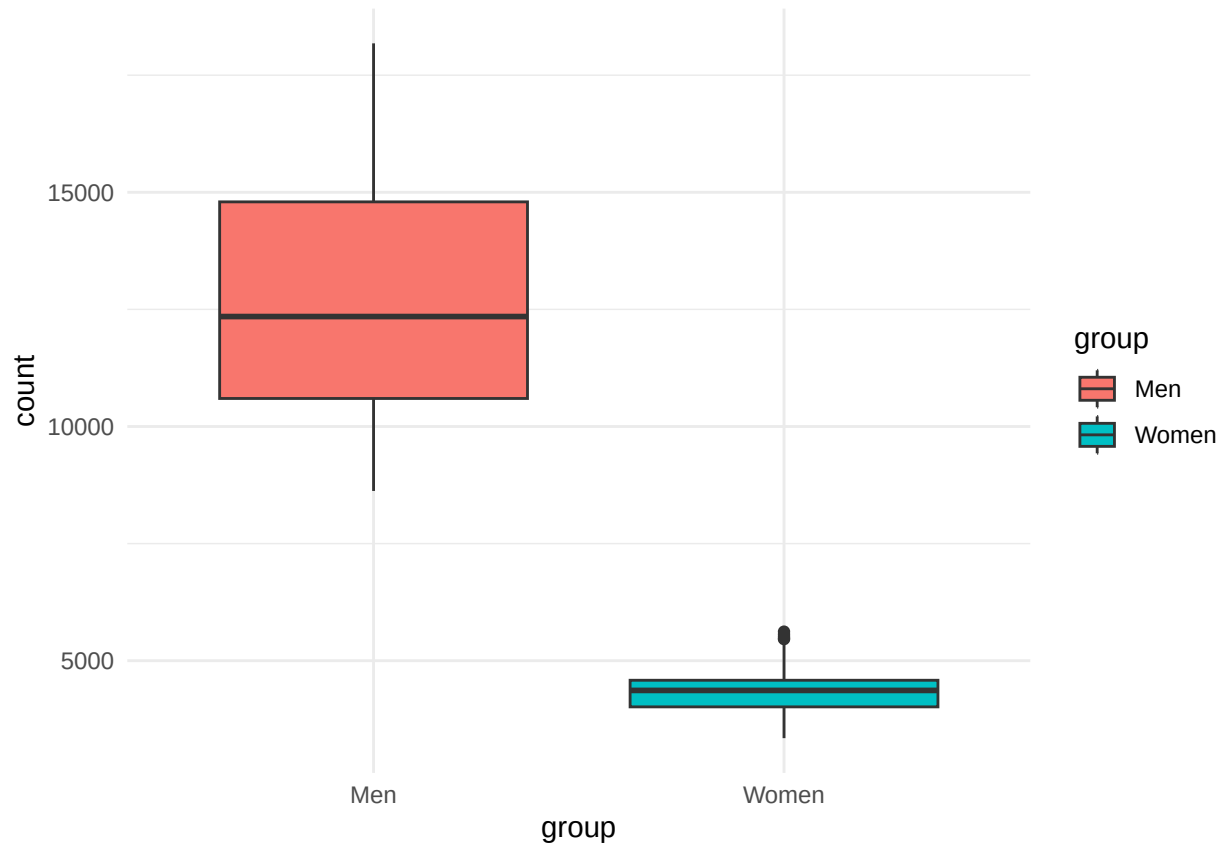
Straight away, there are some key insights we can take from this graph: + There are way more homeless men than there are women. + The male homeless population seems to be rising, while the population of women seems to be a lot more stable.

This could mean a couple things. The box plot below speaks more on this population difference, so I will leave that discussion for that. Here though, I want to discuss why the male population seems to be rising, and more prone to change.

The American Institute for Boys and Men released a [data breakdown](#) as to why homelessness is more likely to happen to men compared to women. For one, men are more likely to be single (or rather, have fewer social bonds) compared to women. Not to mention that when it comes to finding a shelter placement, women are also more likely to find shelter placements compared to men. Now, while there is an obvious gender gap in finding help, it does not mean that there is an implicit “unfair advantage.” Women are more likely to face violence on the streets, so it makes sense that women would be more likely to be accepted into a shelter.

The disparity is also largely driven by veterans that are unable to return to daily life. Unfortunately, veterans in the United States are not given as much support as required for them to re-enter society. I would like to explore this angle further, but for the sake of staying on topic, I am going to continue keeping things brief.

For the dip during covid, its hard to say why that happened, and why it did not affect women so much. Most likely, it comes back from the idea brought up earlier from the source from AIBM: women tend to have more support from the government and informal social support networks as well. So, hypothetically, when the covid pandemic hit, women who would have been homeless without their support network would have been on the streets. Same with the rent spike, it is possible because of the stronger social connections, women were able to find places to live. These are just guesses off the top of my head though.

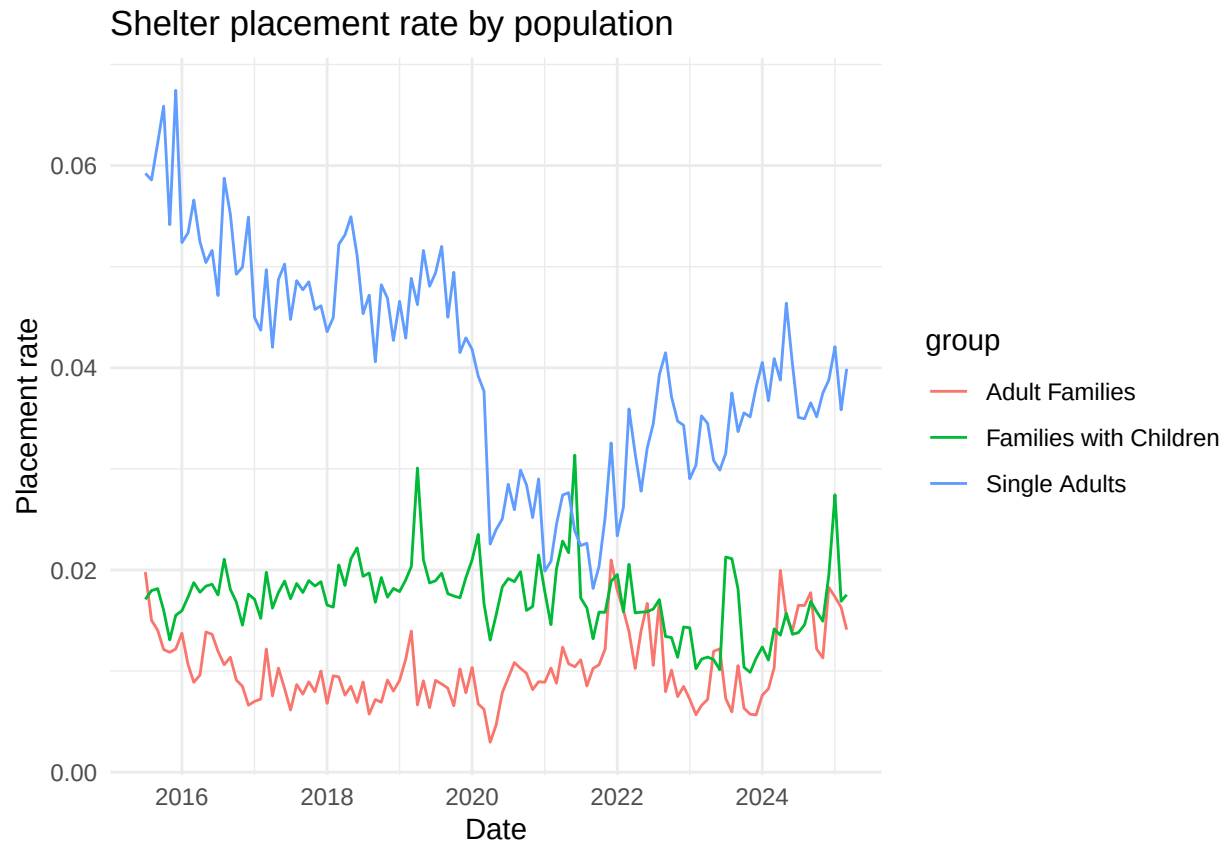


The box plots explain the disparity well. Men have a more variable population in homelessness, while women is more steady, unusually so. The variance for men comes from the fact that the population of homeless men keeps increasing, while women stays constant.

A reason for almost no homeless women could also be that first, this data set comes from Single Adults, not Adult Families or Families with Children. During the migration spike, it could be that the women that did come in were already in a family, therefore classified into a different category. From the same source of AIBM also states that women were more likely to be homeless if they were mothers, which sheds insight as to why they aren't in this category.

Homeless Shelter Placement Rates

Homeless shelter placements can be characterized as housing placements after leaving the shelter. Instead of taking the raw numbers, I will instead be plotting and taking a look at the rate of placements, as this gives us better information as to what is happening.



A couple of things are strikingly obvious when looking at the chart, that is that single adults tend to have higher rates of being accepted than families with children or adult families. But before speaking about this, we should take a look at the actual percentage rates, which make the bigger story. Out of the whole homeless population in NYC, we don't see numbers higher than ~7%. This indicates that New York does not have nearly as much housing capacity for the homeless as needed.

Other observations that can be made in the graph is that Families with Children seems to be more stationary than adult family and single adult populations. Adult families is plausibly stationary from appearance alone, at least much more than single adults. From single adults, we can see a lot more fluctuation in the population of homeless single adults. For single adults, we see a sharp decrease in placements around the same time as the COVID-19 pandemic, and a steady rise as the years go on.

Some guesses as to why single adults tend to be placed at higher rates compared to families could be that single adults tend to come in and out of homelessness, and are easier to accommodate than families. This isn't to say that single adults receive more help than families with children,

Regression Analysis

For this section, I will be running a regressive model for purely analysis. I will not be removing any data points as they are all statistically significant in building our understanding. As such, these models will have heavy skews. My goal in this section is to explore the data, and seeing what are the most influential features in finding shelter placements.

For this regression, we will be applying a Quasi-Poisson regression. It's built on Poisson, but it loosens the assumption that variance must exactly equal the mean, allowing the data to be more spread out than standard Poisson permits. This is elaborated on shortly.

Families with Children Regression Analysis

In this block, we apply a regression on all the data points that involve families with children. The analysis will be below the plots.

Part 0: Preparing the data to be modeled

Since this is a Poisson model, this is the mathematical model being used for the regression, as well as an explanation for each feature:

$$\log(\mathbb{E}[Y_i | \mathbf{x}_i]) = \log(\text{Pop}_i) + \beta_0 + \sum_{m=2}^{12} \beta_m \cdot \text{Month}_{i,m} + \sum_{k=1}^K \gamma_k \cdot x_{i,k} \quad (1)$$

where:

- Y_i is the total number of FWC shelter placements observed in month i , with $Y_i \sim \text{Poisson}(\mu_i)$
- Pop_i is the total FWC population in month i , entered as a fixed-coefficient offset ($\log(\text{Pop}_i)$), so the model estimates the placement *rate* μ_i/Pop_i rather than the raw count
- $\text{Month}_{i,m}$ are indicator variables for calendar month, with January as the reference category
- $x_{i,k}$ are the $k = 1, \dots, K$ compositional predictors (e.g., age-bracket or ethnicity proportions), expressed in percentage points, relative to a dropped reference bracket
- $\beta_0, \beta_m, \gamma_k$ are the estimated log-rate-ratio coefficients

Equivalently, isolating the rate directly:

$$\log\left(\frac{\mu_i}{\text{Pop}_i}\right) = \beta_0 + \sum_{m=2}^{12} \beta_m \cdot \text{Month}_{i,m} + \sum_{k=1}^K \gamma_k \cdot x_{i,k} \quad (2)$$

Variance structure and overdispersion correction. The equations above specify the model's mean structure and are unchanged by the correction described below; only the assumed relationship between the mean and the variance differs. Diagnostic checks (Section~X) indicated substantial overdispersion in the residuals, with an estimated dispersion parameter of $\hat{\phi} \approx 11.8$ — far above the value of 1 assumed by a standard Poisson model. Under standard Poisson regression, the variance is assumed to equal the mean:

$$\text{Var}(Y_i) = \mu_i \quad (3)$$

To correct for this, coefficients were estimated using a quasi-Poisson specification, which relaxes this assumption to allow the variance to scale with an estimated dispersion parameter:

$$\text{Var}(Y_i) = \hat{\phi} \cdot \mu_i \quad (4)$$

This adjustment does not change the point estimates of β_0 , β_m , or γ_k , nor the mean structure specified above — the coefficients reported throughout this section are numerically identical under both specifications. What

changes is the estimated standard error attached to each coefficient, which is scaled by $\sqrt{\hat{\phi}}$ relative to the standard Poisson fit. This in turn affects confidence intervals and significance tests, which are reported using the quasi-Poisson standard errors throughout the remainder of this section.

Some changes were needed to be done to the data to prevent catastrophic VIF scores (we are talking VIF scores of 441!). Below are the changes done to FwC specifically:

- created two separate features for the month and year. This also helps see the effect month by month, to examine any potential seasonality components.
- added the population offset for the regression, as `'glm(family='poisson')'` does not do this for us automatically.
- Changed the numbers from being raw population numbers to being a proportion

++ So because this change was done, additional changes needed to be done to the regression. We have to leave out a variable, so that variable will be the age bracket from 0-5. This is the most vulnerable population, so we should expect every other feature to drop in comparison, along with the age bracket 18-29 being the highest of the peaks compared to the others. This is my hypothesis as I expect young parents to take this category.

$$\log\left(\frac{\mu_i}{\text{Pop}_i}\right) = \beta_0 + \sum_{m=2}^{12} \beta_m \cdot \text{Month}_{i,m} + \sum_{k=1}^7 (\gamma_k - \gamma_{0-5}) x_{i,k} \quad (5)$$

where each retained coefficient $(\gamma_k - \gamma_{21-29})$ represents the effect of age bracket k relative to the omitted 0-5 reference bracket, and $x_{i,k}$ are the seven remaining age-bracket shares (percentage points) for observation i .

- Re-scaled the proportions so instead of being 0-1 its 0-100. This was done because there were some crazy rate ratios that appear. The 0-100 scale makes the numbers make a little more sense.

Control Group: Regression with Month only To begin, we are going to establish the A group of our experiment, which is purely doing a regression on our months. Here, we can also see the changes from January (our intercept), in other words, just how much seasonality our model has.

```
library(dplyr)

fwc_reg_months <- data_feat %>%
  select(c("month",
           "fwc_total_shelter_placements") %>%
         na.omit())

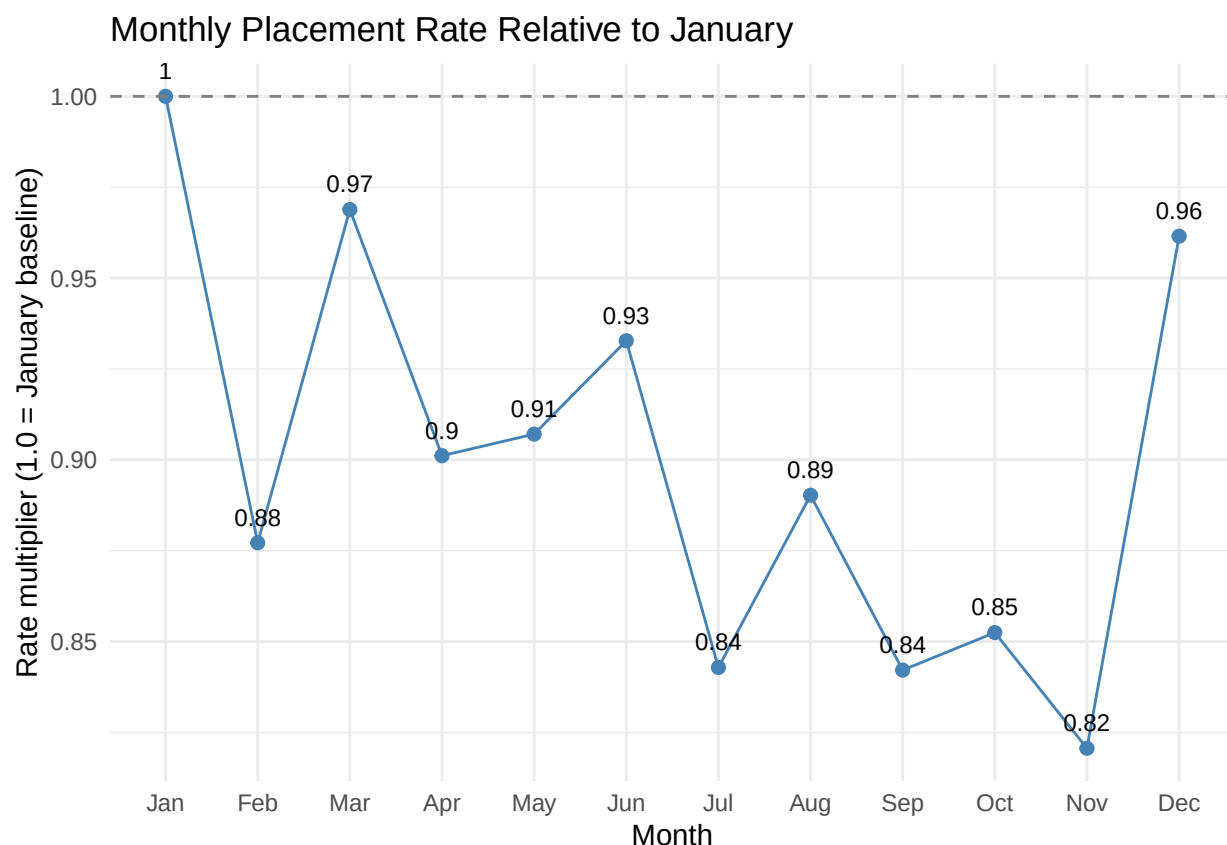
target <- "fwc_total_shelter_placements"

predictors <- setdiff(names(fwc_reg_months), target)

formula <- as.formula(
  paste(target, "~", paste(predictors, collapse = " + "))
)

model_fwc_months <- glm(formula, data = fwc_reg_months, family="quasipoisson")
summary(model_fwc_months)
```

```
##
## Call:
## glm(formula = formula, family = "quasipoisson", data = fwc_reg_months)
##
## Coefficients:
##      Estimate Std. Error t value Pr(>|t|)
## (Intercept)  6.61767    0.08311  79.630 <2e-16 ***
## month2      -0.13105    0.12157  -1.078  0.283
## month3      -0.03164    0.11847  -0.267  0.790
## month4      -0.10411    0.12419  -0.838  0.404
## month5      -0.09754    0.12396  -0.787  0.433
## month6      -0.06961    0.12302  -0.566  0.573
## month7      -0.17094    0.11981  -1.427  0.157
## month8      -0.11624    0.11815  -0.984  0.327
## month9      -0.17181    0.11984  -1.434  0.155
## month10     -0.15962    0.11946  -1.336  0.184
## month11     -0.19768    0.12378  -1.597  0.113
## month12     -0.03925    0.11870  -0.331  0.742
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 51.67422)
##
##      Null deviance: 5728.4  on 120  degrees of freedom
## Residual deviance: 5429.1  on 109  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 4
```



For this next part, this is purely me hypothesizing. We removed a lot of information from the graph, only taking the months into account. Therefore, we are only looking at what happens seasonally for this section.

When we only look at the rate of homeless shelter placements, we see a peak in January (our intercept) compared to the rest of the months. There seems to be a somewhat downtrend until the graph reaches December, where it gets really close to one. The next month being January, this seems to suggest that there is a sharp increase that happens in December that follows into January, resulting in a peak in January. There's a couple of things it could be, but what is happening is that suddenly in these two months there is an increase in the supply in housing. Another way to look at it, is there is a stable decrease in shelter placement. Whether that is by lower demand or lower supply, who's to say? Before investigating, I was of the opinion that it has something to do with leases being renewed, or something involving the school year.

There's also some things to keep in mind, we are looking at placement rates. We should also keep in mind this model takes out a LOT of data from underneath, data that will be further explored in later sections. For now, I did some digging around and I found a couple of explanations that could explain why this is happening:

- [Lease turnover in June is at its highest](#). This coupled with the school year ending, it makes sense that June would be the month that has an increase in placements; probably coincides with shelter placements (or an inability to find an apartment in time, therefore being placed by the city?)
- Data from 2023-2024 shows that evictions peak in May, increasing from November steadily, as shown by this [infographic from EvictionLab](#). This almost perfectly lines up with the seemingly increasing rates we see from December all the way to June.

```

#regression based on shelter placements
fwc_reg_age_ref05 <- data_feat %>%
  select(all_of(c("month", age_cols, "fwc_total_shelter_placements"))) %>%
  na.omit() %>%
  mutate(fwc_total_population = rowSums(across(all_of(age_cols)))) %>%
  mutate(across(all_of(age_cols), ~ 100 * .x / fwc_total_population, .names = "pctpt_{.col}")) %>%
  select(month, starts_with("pctpt_"), fwc_total_population, fwc_total_shelter_placements) %>%
  select(-`pctpt_FWC.Unique.Individuals.by.Age...0.thru.5`) # 0-5 is now the reference

model_ref05 <- glm(
  fwc_total_shelter_placements ~ . - fwc_total_population
  + offset(log(fwc_total_population)),
  data = fwc_reg_age_ref05,
  family = "quasipoisson"
)

summary(model_ref05)

```

Families with Children Age breakdown

```

##
## Call:
## glm(formula = fwc_total_shelter_placements ~ . - fwc_total_population +
##       offset(log(fwc_total_population)), family = "quasipoisson",
##       data = fwc_reg_age_ref05)
##
## Coefficients:
##                                     Estimate Std. Error t value
## (Intercept)                        3.24873    2.50924   1.295
## month2                          -0.14047    0.05823  -2.413
## month3                          -0.04985    0.05705  -0.874
## month4                          -0.05531    0.05995  -0.923
## month5                          -0.02609    0.05974  -0.437
## month6                          -0.01790    0.05936  -0.302
## month7                          -0.10914    0.05920  -1.844
## month8                          -0.05371    0.05711  -0.940
## month9                          -0.11329    0.05773  -1.962
## month10                         -0.11503    0.05766  -1.995
## month11                         -0.18736    0.05924  -3.163
## month12                         -0.03126    0.05687  -0.550
## pctpt_FWC.Unique.Individuals.by.Age...6.thru.13 -0.12518    0.04846  -2.583
## pctpt_FWC.Unique.Individuals.by.Age...14.thru.17 -0.27552    0.11578  -2.380
## pctpt_FWC..Unique.Individuals.by.Age...18.thru.20 -0.03875    0.18996  -0.204
## pctpt_FWC.Unique.Individuals.by.Age...21.thru.29 -0.21200    0.04097  -5.174
## pctpt_FWC.Unique.Individuals.by.Age...30.thru.44 -0.04318    0.02543  -1.698
## pctpt_FWC.Unique.Individuals.by.Age...45.thru.64  0.43879    0.11229   3.908
## pctpt_FWC.Unique.Individuals.by.Age...65.and.above 0.86894    0.99573   0.873
##                                     Pr(>|t|)
## (Intercept)                        0.198345
## month2                            0.017628 *
## month3                            0.384257
## month4                            0.358359

```

```
## month5 0.663205
## month6 0.763631
## month7 0.068140 .
## month8 0.349203
## month9 0.052458 .
## month10 0.048698 *
## month11 0.002060 **
## month12 0.583692
## pctpt_FWC.Unique.Individuals.by.Age...6.thru.13 0.011207 *
## pctpt_FWC.Unique.Individuals.by.Age...14.thru.17 0.019185 *
## pctpt_FWC..Unique.Individuals.by.Age..18.thru.20 0.838745
## pctpt_FWC.Unique.Individuals.by.Age...21.thru.29 1.15e-06 ***
## pctpt_FWC.Unique.Individuals.by.Age...30.thru.44 0.092526 .
## pctpt_FWC.Unique.Individuals.by.Age...45.thru.64 0.000168 ***
## pctpt_FWC.Unique.Individuals.by.Age...65.and.above 0.384892
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 11.80439)
##
## Null deviance: 3958.3 on 120 degrees of freedom
## Residual deviance: 1156.1 on 102 degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 4
```

Using the QuasiPoisson regression, there are a couple of things that stand out, that being the importance of the age groups compared to the months. This is huge, because we have reasonable evidence to assert the claim that shelter placements for families is not affected so much by the season, but by the age groups. The most significant age groups to our model so far seem to be 6-13, 21-29, and 45-64. We do not see the age group of 0-5 as that is included in our intercept, so to speak. The months vary in significance, but they have p values that exceed the usual significance level of 0.05.

Another quick glance before we look into how age plays into this model exactly, lets take a look at the differences in the deviance between the two models. Deviance, by the way, is the “score” of the model. The further we are from zero, the more work the model could use in effectively predicting outcomes. So when we compare the two models, we can see that age is HUGE at explaining homelessness, way more than seasonality ever could. We have a whopping difference of 4,273 points between the models, making age a much better explainer as to what exactly causes homelessness.

So far, we can draw reasonable conclusions about the role of age placing families as opposed to the season. Below is a comparison of both tables:

```
## Warning: package 'knitr' was built under R version 4.5.3
```

Table 1: Model fit comparison: month-only vs. month + age composition

Model	Predictors	Null Deviance	Residual Deviance	Deviance Explained	Residual df	AIC
Month only	month	5728.4	5429.1	1770.1	109	6458.9
Month + Age brackets	month + 8 age brackets	3958.3	1156.1	4273.0	102	2200.0

Something definitely worth checking before we continue, is overdispersion. I strongly suspect we are likely to find a lot of that here, given that FwC has data that (expectedly) moves together, so there is bound to be multicollinearity AND overdispersion. More than likely, the surge in 2022 was a big reason for that surge as well. This is worth looking into. Below,

```
library(dplyr)
library(lubridate)

#full model without taking into account the 2022 surge
model_full <- model_ref05

dispersion_full <- deviance(model_full) / df.residual(model_full)

# Surge-excluded model (June-Dec 2022 removed)

fwc_reg_age_no2022 <- data_feat %>%
  mutate(year = year(Report.Date)) %>% # adjust column name if not Report.Date
  select(all_of(c("year", "month", age_cols, "fwc_total_shelter_placements"))) %>%
  na.omit() %>%
  mutate(fwc_total_population = rowSums(across(all_of(age_cols)))) %>%
  mutate(across(all_of(age_cols), ~ 100 * .x / fwc_total_population, .names = "pctpt_{.col}")) %>%
  filter(!(year == 2022 & month %in% c(6,7,8,9,10,11,12))) %>%
  select(month, starts_with("pctpt_"), fwc_total_population, fwc_total_shelter_placements) %>%
  select(-`pctpt_FWC.Unique.Individuals.by.Age...0.thru.5`)

model_no2022 <- glm(
  fwc_total_shelter_placements ~ . - fwc_total_population
  + offset(log(fwc_total_population)),
  data = fwc_reg_age_no2022,
  family = "poisson"
)

dispersion_no2022 <- deviance(model_no2022) / df.residual(model_no2022)

# 3. Side-by-side summary for the write-up
dispersion_summary <- data.frame(
  Model = c("Full model (all months)", "Surge excluded (Jun-Dec 2022 removed)"),
  Residual_Deviance = c(deviance(model_full), deviance(model_no2022)),
  Residual_df = c(df.residual(model_full), df.residual(model_no2022)),
  Dispersion_Ratio = c(dispersion_full, dispersion_no2022)
)

cat("\n===== Dispersion comparison =====\n")

##
## ===== Dispersion comparison =====

print(dispersion_summary, row.names = FALSE)

##
Model Residual_Deviance Residual_df
```

```
##           Full model (all months)           1156.089           102
## Surge excluded (Jun-Dec 2022 removed)       1127.400           95
## Dispersion_Ratio
##           11.33420
##           11.86737
```

```
cat("\nInterpretation threshold: a dispersion ratio near 1 indicates\n")
```

```
##
## Interpretation threshold: a dispersion ratio near 1 indicates
```

```
cat("variance consistent with the Poisson assumption. Values well\n")
```

```
## variance consistent with the Poisson assumption. Values well
```

```
cat("above 1 indicate overdispersion -- standard errors and p-values\n")
```

```
## above 1 indicate overdispersion -- standard errors and p-values
```

```
cat("from a standard Poisson fit should be treated as optimistic.\n")
```

```
## from a standard Poisson fit should be treated as optimistic.
```

```
cat("\nChange in dispersion after removing the 2022 surge period:\n")
```

```
##
## Change in dispersion after removing the 2022 surge period:
```

```
cat(round(dispersion_full, 2), "->", round(dispersion_no2022, 2),
    " (", ifelse(dispersion_no2022 < dispersion_full, "decreased", "did not decrease"),
    ")\n", sep = "")
```

```
## 11.33->11.87 (did not decrease)
```

Couple of things to note. First, residual deviance barely decreased when we removed June-December, meaning this isn't that much of a meaningful change.

More importantly, the dispersion ratio is crazy. Dispersion is an important term to understand here, before we continue. Dispersion was assessed as $\hat{\phi} = D/(n - p)$, the ratio of residual deviance to residual degrees of freedom, where values well above 1 indicate overdispersion. In other words, is our variance what the model expects, or is it larger (or smaller) than the model assumes? Typically, a value of 1 is perfect, this means our model pretty much nailed the pattern of the data.

That is not what we are seeing here. In fact, we are seeing some catastrophic numbers. This means that there are a lot of red flags with the model, and potentially not the greatest fit for the model.

Now, lets focus on what ages exactly are housed in homeless shelters.

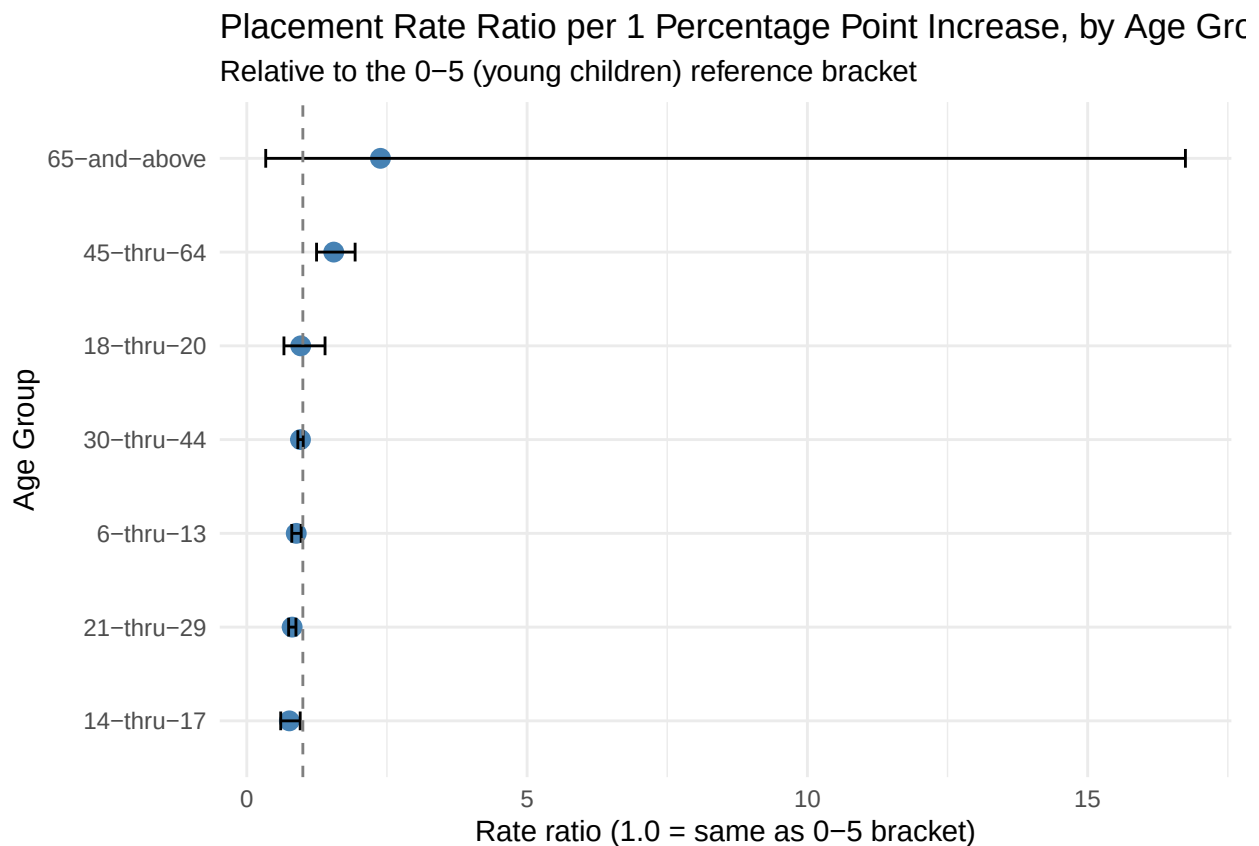
```

library(ggplot2)
library(tidyverse)
library(broom)

age_coefs <- tidy(model_ref05, conf.int = TRUE) %>%
  filter(grepl("^pctpt_FWC", term)) %>%
  mutate(
    term = gsub("pctpt_FWC\\.Unique\\.Individuals\\.by\\.Age\\.+", "", term),
    term = gsub("\\.", "-", term),
    rate_ratio = exp(estimate),
    ci_low = exp(conf.low),
    ci_high = exp(conf.high)
  )

ggplot(age_coefs, aes(x = reorder(term, rate_ratio), y = rate_ratio)) +
  geom_point(size = 3, color = "steelblue") +
  geom_errorbar(aes(ymin = ci_low, ymax = ci_high), width = 0.2) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "gray50") +
  coord_flip() +
  labs(
    title = "Placement Rate Ratio per 1 Percentage Point Increase, by Age Group",
    subtitle = "Relative to the 0-5 (young children) reference bracket",
    x = "Age Group", y = "Rate ratio (1.0 = same as 0-5 bracket)"
  ) +
  theme_minimal()

```



When we compare the age group 0-5 with the rest of the variables, we see some interesting patterns emerge. Firstly, we can see that the most uncertain rate is with 65-and-above. Even with the wide range, it is still the most likely age group to find placement compared to the others. If I had to take a guess as to why, this is because if a family has someone 65+ taking care of a 0-5 year old, I imagine they would need a lot more assistance in finding shelter as they cannot work as much as the other groups.

Some other surprises that show up here is that 21-29 is among the lowest, I did not expect this result. What I DID expect was 14-17 being the lowest. At this age, The children are more independent and are able to roam around more, and perhaps work under the table jobs. Of course, please remember this is all pure conjecture, an educated guess at most. Seeing 6-13 being lower does not surprise me as much, since 6 years old is around the time that children being schooling. Families receive government assistance, and parents are able to work while children are in school.

45-64 being considerably higher is interesting though. Compared to 0-5 year olds, it makes me wonder if older parents are more likely to be placed in housing due to preexisting conditions. Here however, I am not sure.

```
anova(model_ref05, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model: quasipoisson, link: log
##
## Response: fwc_total_shelter_placements
##
## Terms added sequentially (first to last)
##
##
##              Df Deviance Resid. Df
## NULL                      120
## month                    11   274.14   109
## pctpt_FWC.Unique.Individuals.by.Age...6.thru.13    1  1545.96   108
## pctpt_FWC.Unique.Individuals.by.Age...14.thru.17    1   70.08   107
## pctpt_FWC..Unique.Individuals.by.Age..18.thru.20    1    9.92   106
## pctpt_FWC.Unique.Individuals.by.Age...21.thru.29    1   234.52   105
## pctpt_FWC.Unique.Individuals.by.Age...30.thru.44    1   484.44   104
## pctpt_FWC.Unique.Individuals.by.Age...45.thru.64    1   174.22   103
## pctpt_FWC.Unique.Individuals.by.Age...65.and.above  1    8.98   102
##
##              Resid. Dev Pr(>Chi)
## NULL                      3958.3
## month                    3684.2 0.0164339 *
## pctpt_FWC.Unique.Individuals.by.Age...6.thru.13    2138.2 < 2.2e-16 ***
## pctpt_FWC.Unique.Individuals.by.Age...14.thru.17    2068.2 0.0148250 *
## pctpt_FWC..Unique.Individuals.by.Age..18.thru.20    2058.2 0.3593046
## pctpt_FWC.Unique.Individuals.by.Age...21.thru.29    1823.7 8.300e-06 ***
## pctpt_FWC.Unique.Individuals.by.Age...30.thru.44    1339.3 1.492e-10 ***
## pctpt_FWC.Unique.Individuals.by.Age...45.thru.64    1165.1 0.0001222 ***
## pctpt_FWC.Unique.Individuals.by.Age...65.and.above  1156.1 0.3831281
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

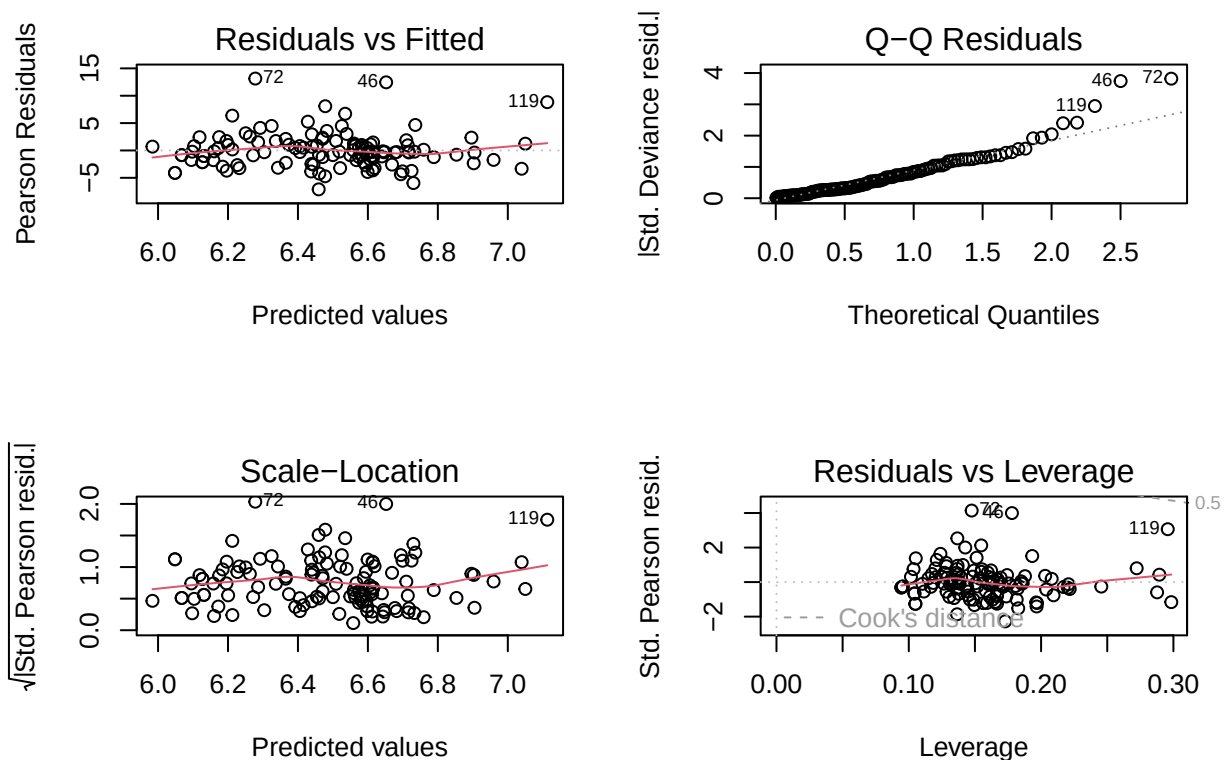
Couple of findings:

The deviance explained from both the age groups 6-13 and 30-44 is enormous, as they scored 1545.96 and 484.44 respectfully. What this potentially tells us, is that families that are most at risk of becoming homeless

are those with dependents. Something else that is quite interesting is 18-20 explaining the least. This is somewhat expected, as there aren't any families that typically have 18-20 year olds as the head of house, or needing shelter placement. As discussed before, it could be family units that have 18-20 year olds don't need the housing in the first place, as they are considered to be the healthiest and are already of working age. There is also a smaller possibility of people at this age bracket needing housing. One final point to add here, is 18-20 is also a legitimately small age bracket, way smaller than the others.

What this ANOVA model tells us, is that most of the families receiving shelter placements are families with children from ages 0-13, and the parents are typically 30-44. The parents can also be 21-29 and 45-64, although this is less likely.

```
par(mfrow=c(2, 2))
plot(model_ref05)
```



#There is a large tail but otherwise normal. Probably the surge in 2022

There are a couple of things that warrant discussion in these graphs. First, we can see some very clear outliers in the Residuals vs Fitted graph.

```
library(car)
```

```
## Loading required package: carData
```

```
##
```

```
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
##
##   recode
```

```
## The following object is masked from 'package:purrr':
##
##   some
```

```
vif(model_ref05)
```

```
##
##                               GVIF Df GVIF^(1/(2*Df))
## month                        1.297901 11      1.011923
## pctpt_FWC.Unique.Individuals.by.Age...6.thru.13 11.807912 1      3.436264
## pctpt_FWC.Unique.Individuals.by.Age...14.thru.17  8.941131 1      2.990172
## pctpt_FWC.Unique.Individuals.by.Age...18.thru.20  3.442212 1      1.855320
## pctpt_FWC.Unique.Individuals.by.Age...21.thru.29  7.288971 1      2.699809
## pctpt_FWC.Unique.Individuals.by.Age...30.thru.44 26.255210 1      5.123984
## pctpt_FWC.Unique.Individuals.by.Age...45.thru.64 14.029413 1      3.745586
## pctpt_FWC.Unique.Individuals.by.Age...65.and.above 3.359532 1      1.832903
```

#Insane multicollinearity, but how to handle this within the parameters of the question trying to be as

As a final sanity check, we do see some multicollinearity with this dataset with some variables. It's a given that a data set like this will have crazy collinearity values, and there isn't much that can be done to reduce these without destroying the whole intent of this analysis. The highest value for VIF here is 30-44 with a value of ~26, well above accepted standards. What is probably happening here is that family placement cycle perfectly correlates with the 30-44 group - the group most likely to be parents, with the children themselves, so of course the VIF is going to skyrocket.

Families with Children Ethnicity Breakdown

```
library(dplyr)
library(car)

eth_cols <- c(
  "FWC.Race.Ethnicity.HoH...Asian.Pacific.Islander",
  "FWC.Race.Ethnicity.HoH...Black.non.Hispanic",
  "FWC.Race.Ethnicity.HoH...Hispanic",
  "FWC.Race.Ethnicity.HoH...Native.American",
  "FWC.Race.Ethnicity.HoH...White.non.Hispanic",
  "FWC.Race.Ethnicity.HoH...Unknown"
)

fwc_reg_eth <- data_feat %>%
  select(all_of(c("month", eth_cols, "fwc_total_shelter_placements"))) %>%
  na.omit() %>%
  mutate(fwc_total_population = rowSums(across(all_of(eth_cols)))) %>%
  mutate(across(all_of(eth_cols), ~ 100 * .x / fwc_total_population, .names = "pctpt_{.col}")) %>%
  select(month, starts_with("pctpt_"), fwc_total_population, fwc_total_shelter_placements) %>%
  select(~pctpt_FWC.Race.Ethnicity.HoH...Black.non.Hispanic) # Black Non-Hispanic is now the reference

target <- "fwc_total_shelter_placements"
predictors <- setdiff(names(fwc_reg_eth), c(target, "fwc_total_population"))
```

```

formula <- as.formula(
  paste(target, "~", paste(predictors, collapse = " + "), "+ offset(log(fwc_total_population))")
)

model_fwc_eth <- glm(formula, data = fwc_reg_eth, family = "poisson")

summary(model_fwc_eth)

```

```

##
## Call:
## glm(formula = formula, family = "poisson", data = fwc_reg_eth)
##
## Coefficients:
##
## (Intercept) -1.2291243 0.1230406
## month2 -0.0873465 0.0169628
## month3 0.0178057 0.0165568
## month4 -0.0061637 0.0173863
## month5 0.0020197 0.0173291
## month6 0.0316598 0.0172092
## month7 -0.0380549 0.0169365
## month8 -0.0150477 0.0166478
## month9 -0.1167022 0.0167066
## month10 -0.1088910 0.0166912
## month11 -0.1921941 0.0172256
## month12 -0.0282587 0.0165262
## pctpt_FWC.Race.Ethnicity.HoH...Asian.Pacific.Islander 0.3798232 0.0485186
## pctpt_FWC.Race.Ethnicity.HoH...Hispanic -0.0190207 0.0008618
## pctpt_FWC.Race.Ethnicity.HoH...Native.American -1.1334276 0.1135515
## pctpt_FWC.Race.Ethnicity.HoH...White.non.Hispanic -0.3577898 0.0222408
## pctpt_FWC.Race.Ethnicity.HoH...Unknown 0.0526526 0.0077075
##
## z value Pr(>|z|)
## (Intercept) -9.990 < 2e-16 ***
## month2 -5.149 2.61e-07 ***
## month3 1.075 0.2822
## month4 -0.355 0.7230
## month5 0.117 0.9072
## month6 1.840 0.0658 .
## month7 -2.247 0.0246 *
## month8 -0.904 0.3661
## month9 -6.985 2.84e-12 ***
## month10 -6.524 6.85e-11 ***
## month11 -11.157 < 2e-16 ***
## month12 -1.710 0.0873 .
## pctpt_FWC.Race.Ethnicity.HoH...Asian.Pacific.Islander 7.828 4.94e-15 ***
## pctpt_FWC.Race.Ethnicity.HoH...Hispanic -22.071 < 2e-16 ***
## pctpt_FWC.Race.Ethnicity.HoH...Native.American -9.982 < 2e-16 ***
## pctpt_FWC.Race.Ethnicity.HoH...White.non.Hispanic -16.087 < 2e-16 ***
## pctpt_FWC.Race.Ethnicity.HoH...Unknown 6.831 8.41e-12 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)

```

```
##
## Null deviance: 3491.1 on 120 degrees of freedom
## Residual deviance: 1391.0 on 104 degrees of freedom
## AIC: 2430.9
##
## Number of Fisher Scoring iterations: 4
```

Once again, we see that ethnicity also plays a huge role in explaining the model, way more than each individual month (ie, seasonality) can.

Table 2: Model fit comparison: month-only vs. month + ethnicity composition

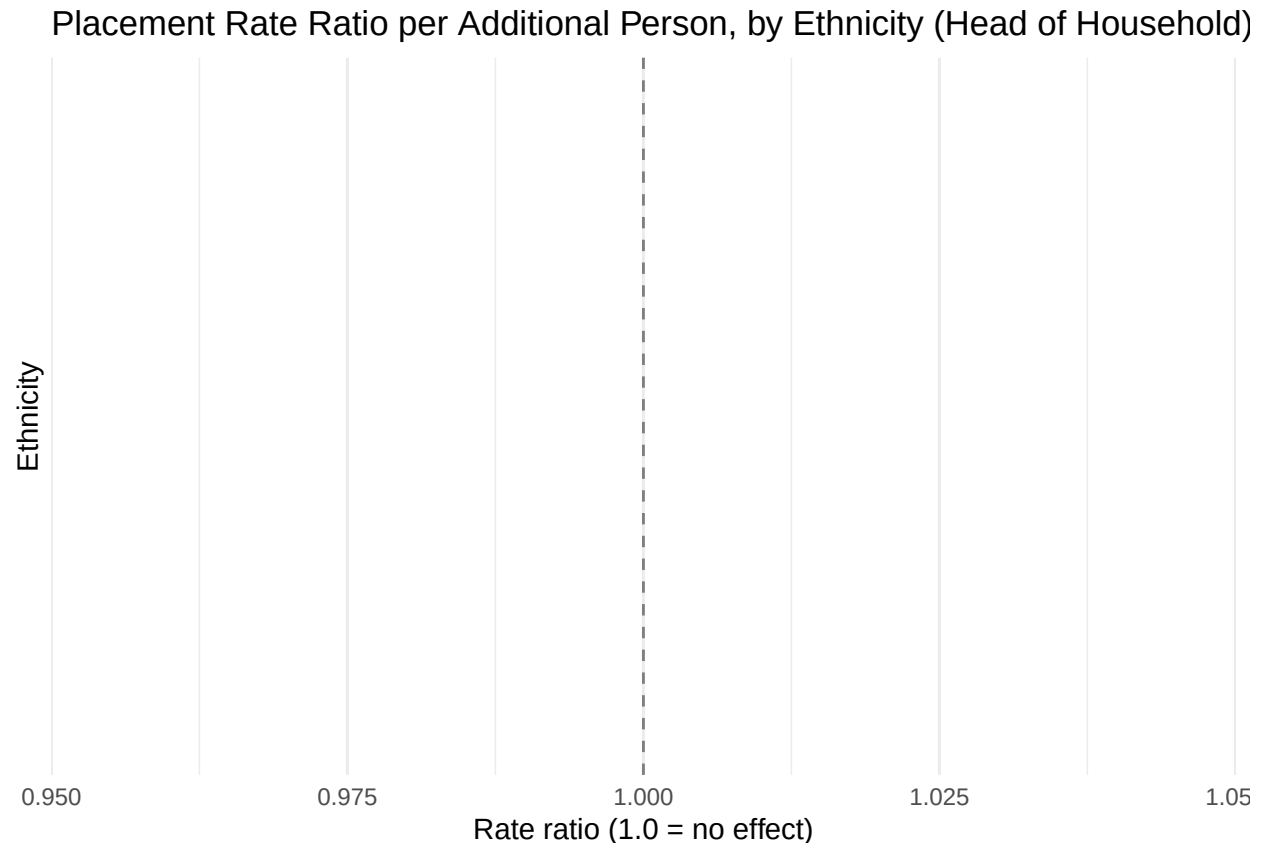
Model	Predictors	Null Deviance	Residual Deviance	Deviance Explained	Residual df	AIC
Month only	month	5728.4	5429.1	299.3	109	6458.9
Month + Age brackets	month + 8 age brackets	5728.4	1101.3	4627.1	103	2053.2

Now, lets focus on what ethnicities exactly are housed in homeless shelters.

```
library(ggplot2)
library(broom)
library(dplyr)

eth_coefs <- tidy(model_fw_c_eth, conf.int = TRUE) %>%
  filter(grepl("^FWC.Race.Ethnicity.HoH", term)) %>%
  mutate(
    term = gsub("FWC.Race.Ethnicity.HoH\\.", "", term),
    term = gsub("\\.", " ", term),
    rate_ratio = exp(estimate),
    ci_low = exp(conf.low),
    ci_high = exp(conf.high)
  )

ggplot(eth_coefs, aes(x = reorder(term, rate_ratio), y = rate_ratio)) +
  geom_point(size = 3, color = "#D2691E") +
  geom_errorbar(aes(ymin = ci_low, ymax = ci_high), width = 0.2, color = "#D2691E") +
  geom_hline(yintercept = 1, linetype = "dashed", color = "gray50") +
  coord_flip() +
  labs(
    title = "Placement Rate Ratio per Additional Person, by Ethnicity (Head of Household)",
    x = "Ethnicity", y = "Rate ratio (1.0 = no effect)"
  ) +
  theme_minimal()
```



Looking at this graph, we can see some interesting patterns. First, Native Americans (~0.6%) are slightly more likely to be placed into a shelter compared to other ethnicities, followed by Asian Pacific Islanders (0.3%). Black Non-Hispanic and Hispanic stay close to the mean rate, while White Non-Hispanics are the least likely to be placed. From the graphs we saw above, these results are surprising as I expected to see Black non-Hispanic and Hispanic to score higher, but do keep in mind we are working with rates.

One hypothesis, consistent with broader literature on structural inequality in housing access (see earlier discussion), is that white households may have relatively more alternative housing options — though this single coefficient, given the model’s high collinearity, is weak evidence on its own and would need to be tested with a model that isolates ethnicity effects more cleanly.

Adult Families Regression Analysis

As discussed before, Adult Families are family units where every member is above 18 years old. Naturally, the ages 0-17 are excluded from these regressions.

The same “control” regression will be used as with FwC, only looking at the coefficients month by month.

```
af_reg_months <- data_feat %>%
  select(c("month",
           "af_total_shelter_placements") %>%
         na.omit())

target <- "af_total_shelter_placements"

predictors <- setdiff(names(af_reg_months), target)
```

```
formula <- as.formula(
  paste(target, "~", paste(predictors, collapse = " + "))
)

model_af_months <- glm(formula, data = af_reg_months, family="poisson")
summary(model_af_months)
```

```
##
## Call:
## glm(formula = formula, family = "poisson", data = af_reg_months)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  3.871201   0.045644  84.814 <2e-16 ***
## month2      -0.031749   0.065068  -0.488  0.6256
## month3       0.002081   0.064516   0.032  0.9743
## month4      -0.130889   0.068716  -1.905  0.0568 .
## month5       0.049672   0.065467   0.759  0.4480
## month6       0.013793   0.066079   0.209  0.8347
## month7       0.020619   0.062758   0.329  0.7425
## month8      -0.048506   0.063808  -0.760  0.4471
## month9      -0.040585   0.063685  -0.637  0.5239
## month10     -0.137870   0.065248  -2.113  0.0346 *
## month11     -0.114663   0.066481  -1.725  0.0846 .
## month12     -0.008368   0.064685  -0.129  0.8971
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 441.85  on 120  degrees of freedom
## Residual deviance: 422.91  on 109  degrees of freedom
## AIC: 1129.2
##
## Number of Fisher Scoring iterations: 4
```

Interestingly, we see that our P values are not as significant. This potentially means that the cycle of placements that was discussed earlier relating to the school year for FwC does not affect Adult Families as much.

```
library(ggplot2)
library(dplyr)

# Build a coefficient summary with real month labels
month_labels <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                  "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")

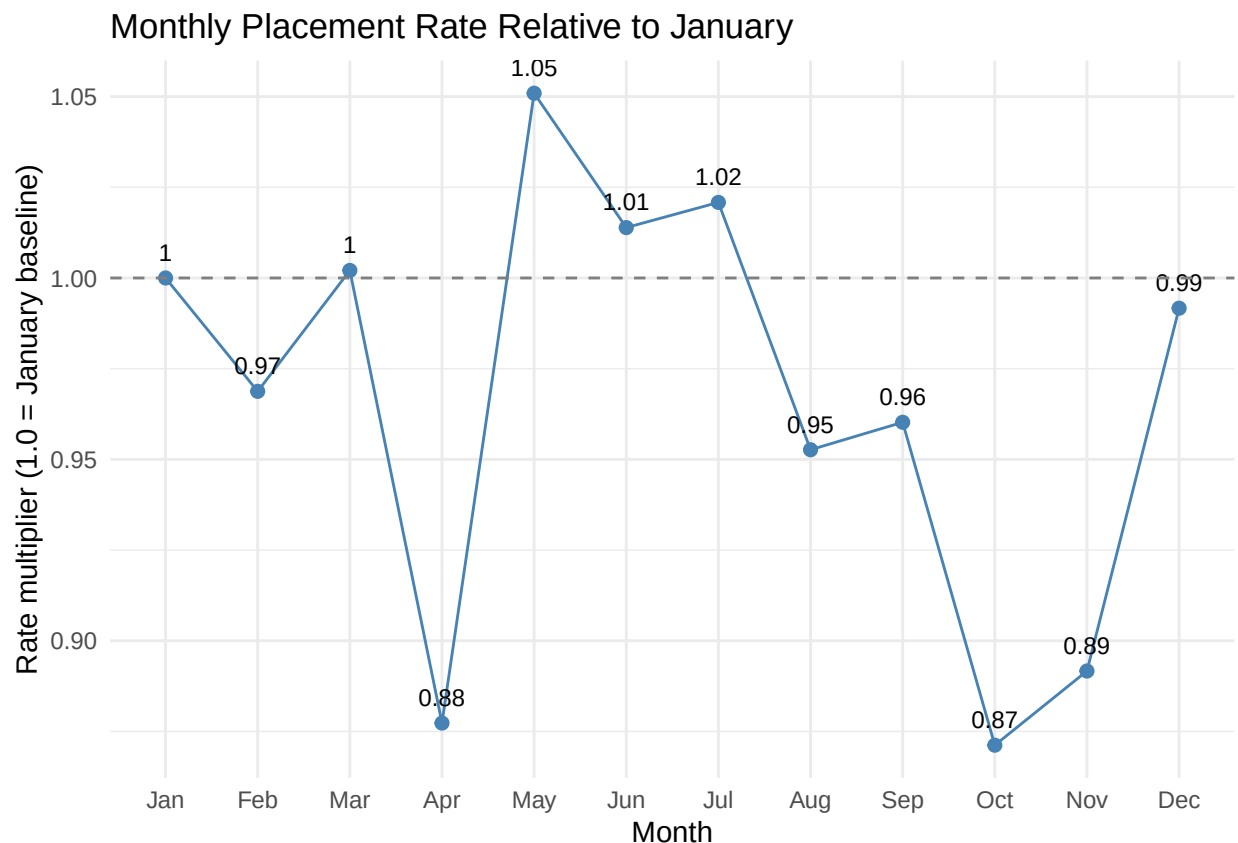
coef_df <- data.frame(
  month = 1:12,
  log_effect = c(0, coef(model_af_months)[grep("^month", names(coef(model_af_months))]))
) %>%
  mutate(
```

```

    month_label = factor(month_labels[month], levels = month_labels),
    rate_multiplier = exp(log_effect)
  )

ggplot(coef_df, aes(x = month_label, y = rate_multiplier, group = 1)) +
  geom_line(color = "steelblue") +
  geom_point(size = 2, color = "steelblue") +
  geom_text(aes(label = round(rate_multiplier, 2)), vjust = -1, size = 3) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "gray50") +
  labs(
    title = "Monthly Placement Rate Relative to January",
    x = "Month", y = "Rate multiplier (1.0 = January baseline)"
  ) +
  theme_minimal()

```



Taking a look at the graph itself though, we

Adult Families Age breakdown

```

#regression based on shelter placements
af_reg_age <- data_feat %>%
  select(c("month",
    "AF.Unique.Individuals.by.Age...18.thru.20",
    "AF.Unique.Individuals.by.Age...21.thru.29",
    "AF.Unique.Individuals.by.Age...30.thru.44",
    "AF.Unique.Individuals.by.Age...45.thru.64",

```

```

      "AF.Unique.Individuals.by.Age...65.and.above",
      "af_total_shelter_placements") %>%
      na.omit())

target <- "af_total_shelter_placements"

predictors <- setdiff(names(af_reg_age), target)

formula <- as.formula(
  paste(target, "~", paste(predictors, collapse = " + "))
)

model_af_age <- glm(formula, data = af_reg_age, family="poisson")
summary(model_af_age)

```

```

##
## Call:
## glm(formula = formula, family = "poisson", data = af_reg_age)
##
## Coefficients:
##
##              Estimate Std. Error z value
## (Intercept)      3.759e+00  1.123e-01 33.460
## month2          -3.196e-02  6.527e-02  -0.490
## month3          -1.549e-03  6.491e-02  -0.024
## month4          -1.045e-01  6.945e-02  -1.504
## month5           6.787e-02  6.556e-02   1.035
## month6           2.935e-02  6.627e-02   0.443
## month7           2.031e-02  6.342e-02   0.320
## month8          -3.647e-02  6.485e-02  -0.562
## month9          -2.479e-02  6.433e-02  -0.385
## month10         -1.176e-01  6.554e-02  -1.794
## month11         -1.004e-01  6.667e-02  -1.505
## month12         -4.783e-03  6.476e-02  -0.074
## AF.Unique.Individuals.by.Age...18.thru.20  1.932e-03  2.364e-04   8.172
## AF.Unique.Individuals.by.Age...21.thru.29  1.540e-04  2.175e-04   0.708
## AF.Unique.Individuals.by.Age...30.thru.44  -6.569e-04  2.932e-04  -2.241
## AF.Unique.Individuals.by.Age...45.thru.64   9.031e-05  5.433e-05   1.662
## AF.Unique.Individuals.by.Age...65.and.above -2.366e-03  5.697e-04  -4.153
##
##              Pr(>|z|)
## (Intercept)    < 2e-16 ***
## month2         0.6244
## month3         0.9810
## month4         0.1325
## month5         0.3006
## month6         0.6579
## month7         0.7487
## month8         0.5739
## month9         0.6999
## month10        0.0728 .
## month11        0.1323
## month12        0.9411
## AF.Unique.Individuals.by.Age...18.thru.20  3.02e-16 ***
## AF.Unique.Individuals.by.Age...21.thru.29   0.4791

```

```
## AF.Unique.Individuals.by.Age...30.thru.44      0.0251 *
## AF.Unique.Individuals.by.Age...45.thru.64      0.0965 .
## AF.Unique.Individuals.by.Age...65.and.above 3.29e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 441.85  on 120  degrees of freedom
## Residual deviance: 347.86  on 104  degrees of freedom
## AIC: 1064.1
##
## Number of Fisher Scoring iterations: 4
```

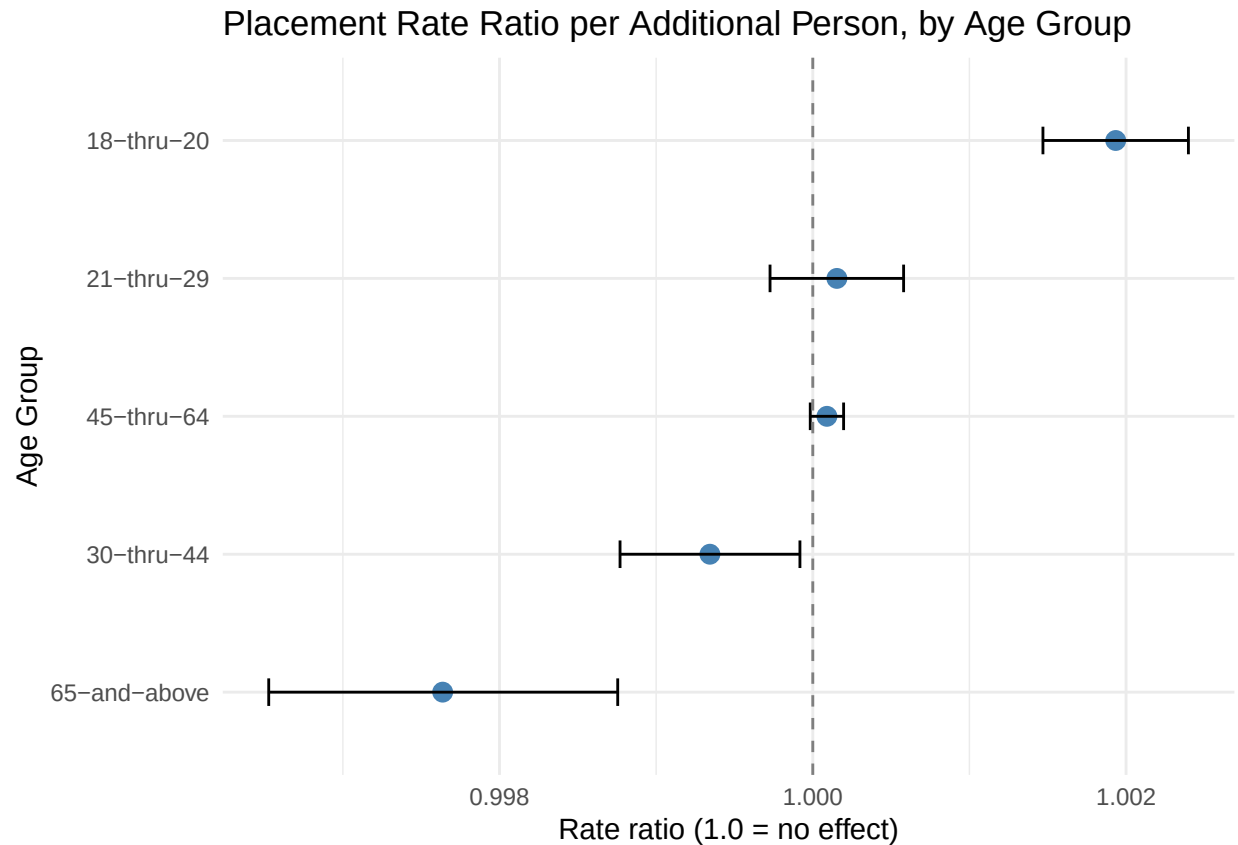
Table 3: Model fit comparison: month-only vs. month + age composition

Model	Predictors	Null Deviance	Residual Deviance	Deviance Explained	Residual df	AIC
Month only	month	5728.4	5429.1	299.3	109	6458.9
Month + Age brackets	month + 8 age brackets	5728.4	1101.2	4627.2	101	2147.1

```
library(ggplot2)
library(tidyverse)
library(broom)

age_coefs <- tidy(model_af_age, conf.int = TRUE) %>%
  filter(grepl("^AF.Unique.Individuals.by.Age", term)) %>%
  mutate(
    term = gsub("AF.Unique.Individuals.by.Age\\\\.", "", term),
    term = gsub("\\\\.", "-", term),
    rate_ratio = exp(estimate),
    ci_low = exp(conf.low),
    ci_high = exp(conf.high)
  )

ggplot(age_coefs, aes(x = reorder(term, rate_ratio), y = rate_ratio)) +
  geom_point(size = 3, color = "steelblue") +
  geom_errorbar(aes(ymin = ci_low, ymax = ci_high), width = 0.2) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "gray50") +
  coord_flip() +
  labs(
    title = "Placement Rate Ratio per Additional Person, by Age Group",
    x = "Age Group", y = "Rate ratio (1.0 = no effect)"
  ) +
  theme_minimal()
```

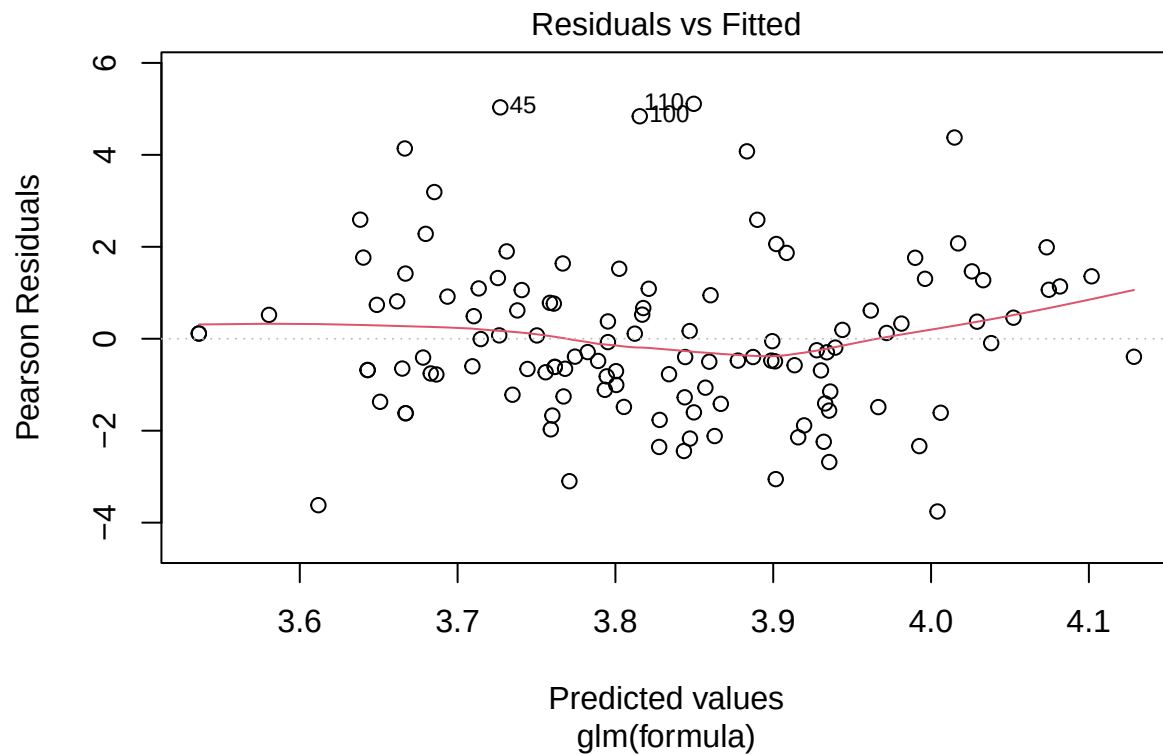


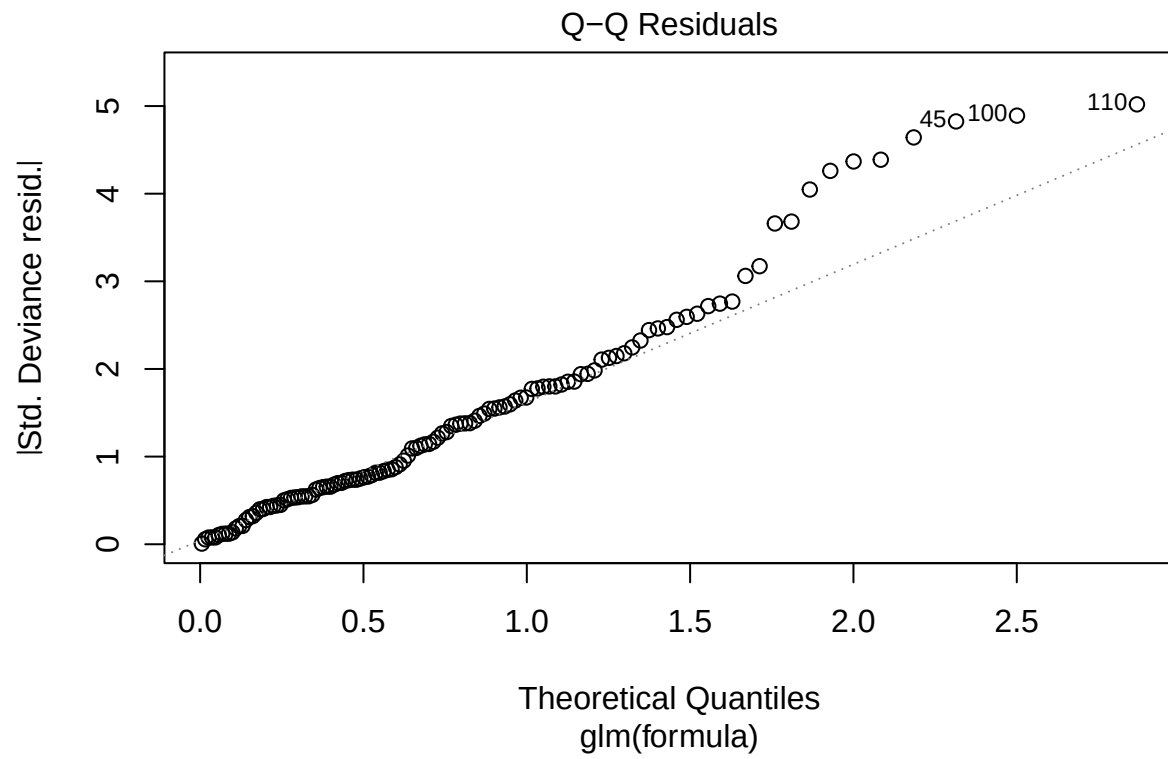
```
anova(model_af_age, test = "Chisq")
```

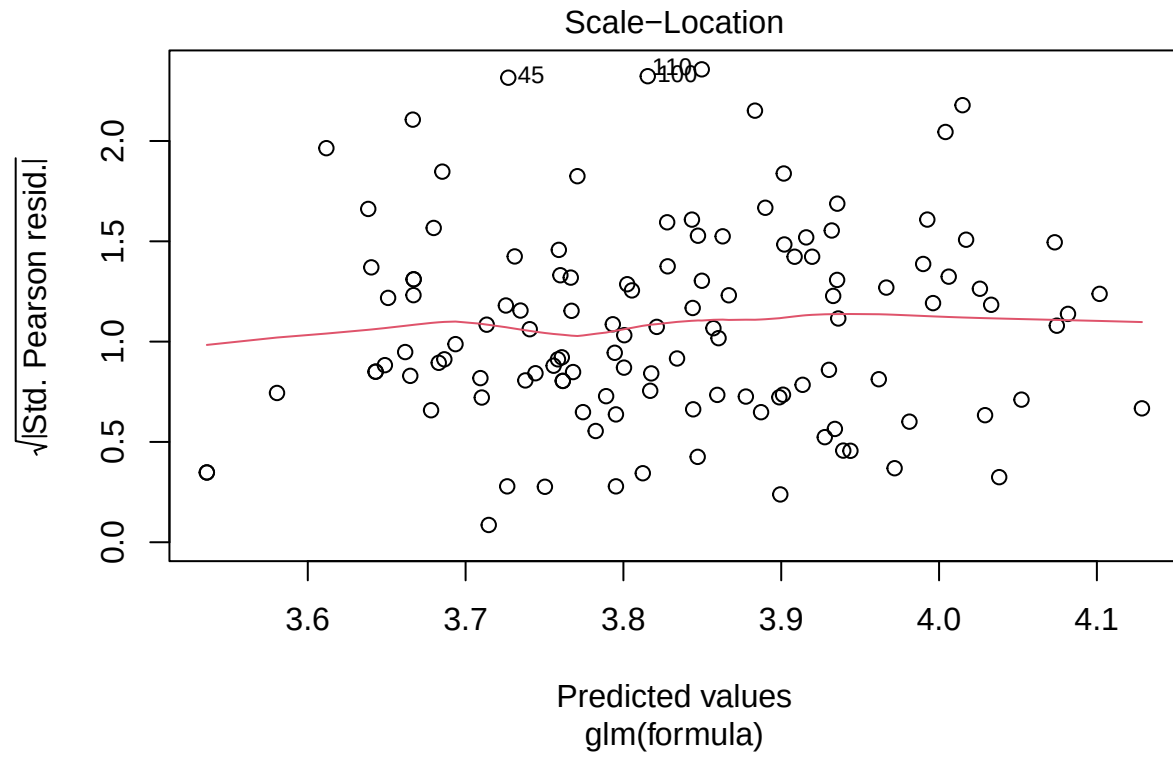
```
## Analysis of Deviance Table
##
## Model: poisson, link: log
##
## Response: af_total_shelter_placements
##
## Terms added sequentially (first to last)
##
##
##          Df Deviance Resid. Df Resid. Dev
## NULL                                120    441.85
## month                                11    18.9371
## AF.Unique.Individuals.by.Age...18.thru.20  1    22.5921
## AF.Unique.Individuals.by.Age...21.thru.29  1     7.2119
## AF.Unique.Individuals.by.Age...30.thru.44  1    18.7347
## AF.Unique.Individuals.by.Age...45.thru.64  1     9.4590
## AF.Unique.Individuals.by.Age...65.and.above 1    17.0469
##          Pr(>Chi)
## NULL
## month          0.062232 .
## AF.Unique.Individuals.by.Age...18.thru.20 2.003e-06 ***
## AF.Unique.Individuals.by.Age...21.thru.29  0.007242 **
## AF.Unique.Individuals.by.Age...30.thru.44  1.502e-05 ***
```

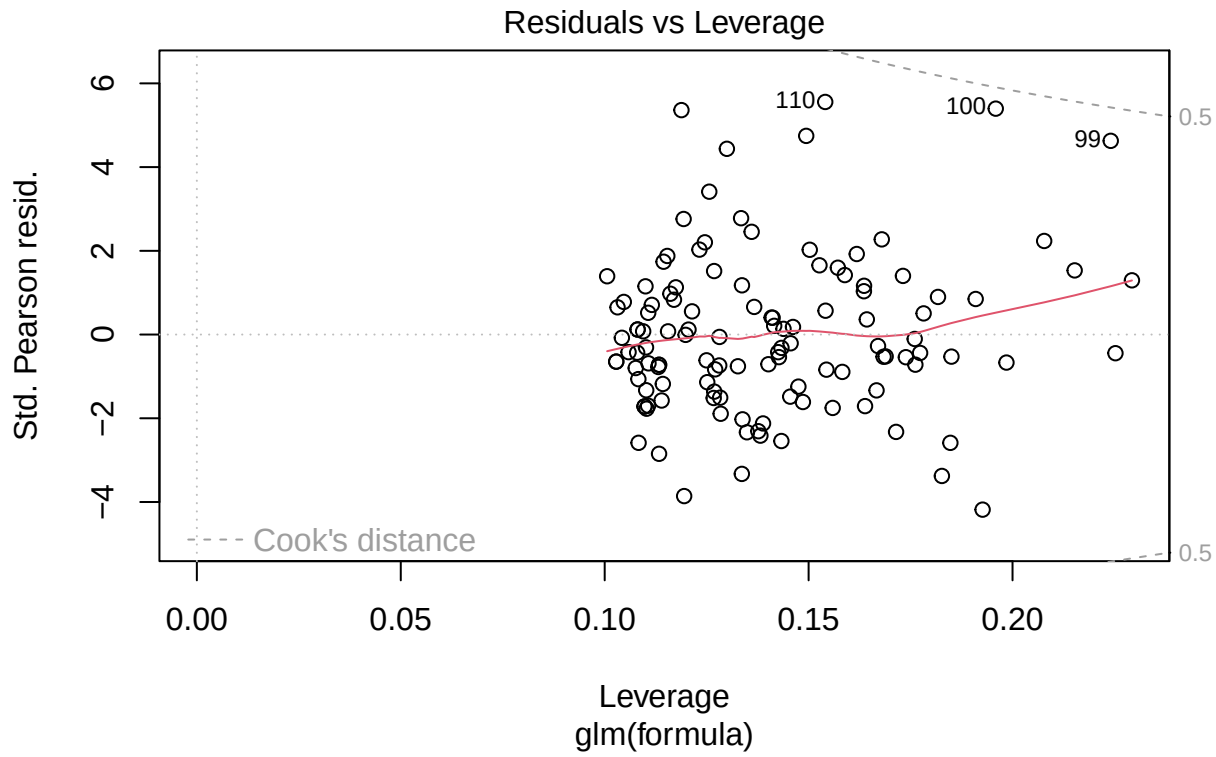
```
## AF.Unique.Individuals.by.Age...45.thru.64    0.002101 **
## AF.Unique.Individuals.by.Age...65.and.above 3.647e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
plot(model_af_age)
```









```
library(car)
vif(model_af_age)
```

##		GVIF	Df	GVIF ^{1/(2*Df)}
##	month	1.079495	11	1.003483
##	AF.Unique.Individuals.by.Age...18.thru.20	4.264874	1	2.065157
##	AF.Unique.Individuals.by.Age...21.thru.29	39.462209	1	6.281895
##	AF.Unique.Individuals.by.Age...30.thru.44	41.779376	1	6.463697
##	AF.Unique.Individuals.by.Age...45.thru.64	2.226732	1	1.492224
##	AF.Unique.Individuals.by.Age...65.and.above	3.133693	1	1.770224

Adult Families Ethnicity Breakdown

```
af_reg_eth <- data_feat %>%
  select(c("month",
    "AF.Race.Ethnicity.HoH...Asian.Pacific.Islander",
    "AF.Race.Ethnicity.HoH...Black.non.Hispanic",
    "AF.Race.Ethnicity.HoH...Hispanic",
    "AF.Race.Ethnicity.HoH...Native.American",
    "AF.Race.Ethnicity.HoH...White.non.Hispanic",
    "AF.Race.Ethnicity.HoH...Unknown",
    "af_total_shelter_placements") %>%
    na.omit())

target <- "af_total_shelter_placements"
```

```

predictors <- setdiff(names(af_reg_eth), target)

formula <- as.formula(
  paste(target, "~", paste(predictors, collapse = " + "))
)

model_af_eth <- glm(formula, data = af_reg_eth, family="poisson")
summary(model_af_eth)

```

```

##
## Call:
## glm(formula = formula, family = "poisson", data = af_reg_eth)
##
## Coefficients:
##
##              Estimate Std. Error z value
## (Intercept)    3.698e+00  1.150e-01  32.163
## month2        -4.829e-02  6.517e-02  -0.741
## month3        -3.000e-02  6.495e-02  -0.462
## month4        -1.790e-01  6.991e-02  -2.561
## month5        -3.187e-02  6.841e-02  -0.466
## month6        -3.603e-02  6.801e-02  -0.530
## month7        -3.907e-02  6.449e-02  -0.606
## month8        -1.073e-01  6.560e-02  -1.635
## month9        -8.204e-02  6.476e-02  -1.267
## month10       -1.402e-01  6.563e-02  -2.136
## month11       -7.798e-02  6.693e-02  -1.165
## month12        4.212e-02  6.555e-02   0.643
## AF.Race.Ethnicity.HoH...Asian.Pacific.Islander -1.217e-03  3.789e-03  -0.321
## AF.Race.Ethnicity.HoH...Black.non.Hispanic    4.111e-04  9.299e-05   4.421
## AF.Race.Ethnicity.HoH...Hispanic              4.690e-05  6.599e-05   0.711
## AF.Race.Ethnicity.HoH...Native.American      -5.568e-02  1.080e-02  -5.155
## AF.Race.Ethnicity.HoH...White.non.Hispanic    4.540e-04  7.376e-04   0.616
## AF.Race.Ethnicity.HoH...Unknown              -8.644e-04  8.383e-04  -1.031
##
##              Pr(>|z|)
## (Intercept)    < 2e-16 ***
## month2          0.4587
## month3          0.6442
## month4          0.0104 *
## month5          0.6413
## month6          0.5963
## month7          0.5446
## month8          0.1020
## month9          0.2052
## month10         0.0327 *
## month11         0.2440
## month12         0.5205
## AF.Race.Ethnicity.HoH...Asian.Pacific.Islander 0.7480
## AF.Race.Ethnicity.HoH...Black.non.Hispanic    9.80e-06 ***
## AF.Race.Ethnicity.HoH...Hispanic              0.4773
## AF.Race.Ethnicity.HoH...Native.American      2.53e-07 ***
## AF.Race.Ethnicity.HoH...White.non.Hispanic    0.5382
## AF.Race.Ethnicity.HoH...Unknown              0.3025
## ---

```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 441.85  on 120  degrees of freedom
## Residual deviance: 372.04  on 103  degrees of freedom
## AIC: 1090.3
##
## Number of Fisher Scoring iterations: 4
```

Table 4: Model fit comparison: month-only vs. month + ethnicity composition

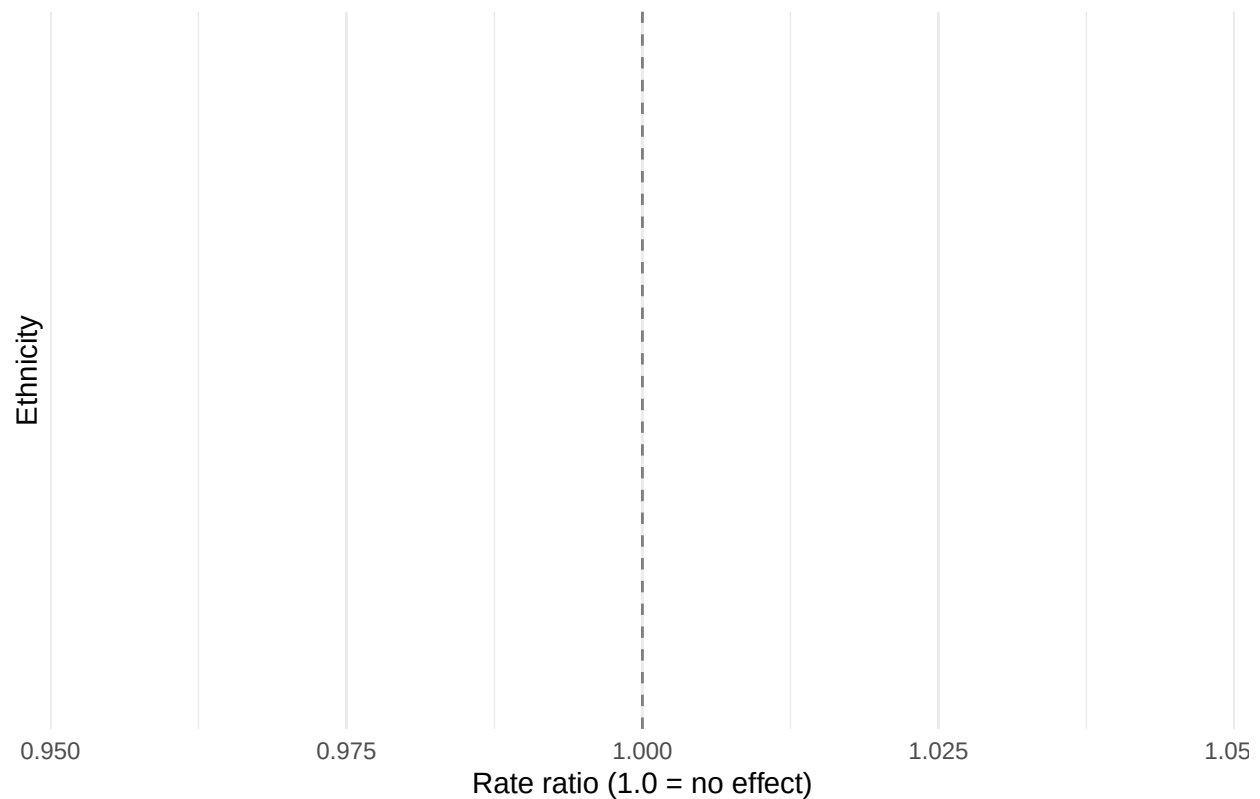
Model	Predictors	Null Deviance	Residual Deviance	Deviance Explained	Residual df	AIC
Month only	month	5728.4	5429.1	299.3	109	6458.9
Month + Age brackets	month + 8 age brackets	5728.4	1101.3	4627.1	103	2053.2

```
library(ggplot2)
library(broom)
library(dplyr)

eth_coefs <- tidy(model_fwc_eth, conf.int = TRUE) %>%
  filter(grepl("^AF.Race.Ethnicity.HoH", term)) %>%
  mutate(
    term = gsub("AF.Race.Ethnicity.HoH\\.", "", term),
    term = gsub("\\.", " ", term),
    rate_ratio = exp(estimate),
    ci_low = exp(conf.low),
    ci_high = exp(conf.high)
  )

ggplot(eth_coefs, aes(x = reorder(term, rate_ratio), y = rate_ratio)) +
  geom_point(size = 3, color = "#D2691E") +
  geom_errorbar(aes(ymin = ci_low, ymax = ci_high), width = 0.2, color = "#D2691E") +
  geom_hline(yintercept = 1, linetype = "dashed", color = "gray50") +
  coord_flip() +
  labs(
    title = "Placement Rate Ratio per Additional Person, by Ethnicity (Head of Household)",
    x = "Ethnicity", y = "Rate ratio (1.0 = no effect)"
  ) +
  theme_minimal()
```

Placement Rate Ratio per Additional Person, by Ethnicity (Head of Household)



Single Adults Regression Analysis

```
sa_reg <- data %>%
  select(starts_with("SA"), sa_total_shelter_placements) %>%
  mutate(across(everything(), as.numeric)) %>%
  na.omit()

target <- "sa_total_shelter_placements"

predictors <- setdiff(names(sa_reg), target)

formula <- as.formula(
  paste(target, "~", paste(predictors, collapse = " + "))
)

model_sa <- glm(formula, data = sa_reg, family = "poisson")
```

```
sa_reg_months <- data_feat %>%
  select(c("month",
           "sa_total_shelter_placements")) %>%
  na.omit()

target <- "sa_total_shelter_placements"
```

```

predictors <- setdiff(names(sa_reg_months), target)

formula <- as.formula(
  paste(target, "~", paste(predictors, collapse = " + "))
)

model_sa_months <- glm(formula, data = sa_reg_months, family="poisson")
summary(model_sa_months)

```

```

##
## Call:
## glm(formula = formula, family = "poisson", data = sa_reg_months)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  6.479277   0.012389  522.978 < 2e-16 ***
## month2      -0.024551   0.017629   -1.393  0.16374
## month3       0.096356   0.017114    5.630 1.80e-08 ***
## month4      -0.010544   0.018051   -0.584  0.55914
## month5       0.028829   0.017866    1.614  0.10661
## month6      -0.012614   0.018061   -0.698  0.48491
## month7      -0.080078   0.017455  -4.588 4.48e-06 ***
## month8      -0.010732   0.017162   -0.625  0.53175
## month9      -0.048679   0.017320  -2.811  0.00495 **
## month10     -0.005527   0.017141   -0.322  0.74711
## month11      0.010535   0.017475    0.603  0.54659
## month12      0.084014   0.017164    4.895 9.85e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 4421.1  on 120  degrees of freedom
## Residual deviance: 4236.1  on 109  degrees of freedom
## AIC: 5263
##
## Number of Fisher Scoring iterations: 4

```

```

library(ggplot2)
library(dplyr)

# Build a coefficient summary with real month labels
month_labels <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                  "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")

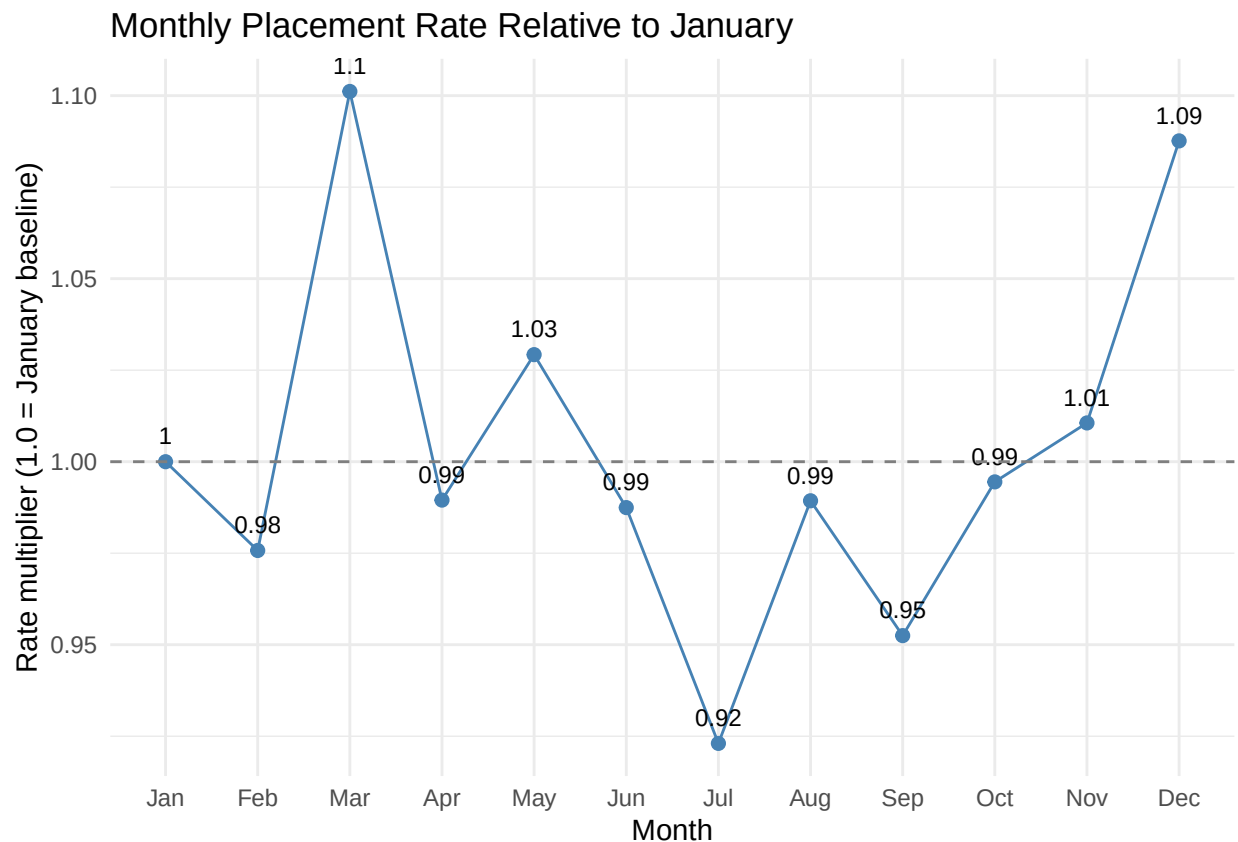
coef_df <- data.frame(
  month = 1:12,
  log_effect = c(0, coef(model_sa_months)[grep("^month", names(coef(model_sa_months)))])
) %>%
  mutate(
    month_label = factor(month_labels[month], levels = month_labels),
    rate_multiplier = exp(log_effect)
  )

```

```
)

ggplot(coef_df, aes(x = month_label, y = rate_multiplier, group = 1)) +
  geom_line(color = "steelblue") +
  geom_point(size = 2, color = "steelblue") +
  geom_text(aes(label = round(rate_multiplier, 2)), vjust = -1, size = 3) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "gray50") +
  labs(
    title = "Monthly Placement Rate Relative to January",
    x = "Month", y = "Rate multiplier (1.0 = January baseline)"
  ) +
  theme_minimal()

```



Single Adult Age breakdown

```
#regression based on shelter placements
sa_reg_age <- data_feat %>%
  select(c("month",
    "SA.Unique.Individuals.by.Age...18.thru.29",
    "SA.Unique.Individuals.by.Age...30.thru.44",
    "SA.Unique.Individuals.by.Age...45.thru.64",
    "SA.Unique.Individuals.by.Age...65.and.above",
    "sa_total_shelter_placements") %>%
    na.omit())

target <- "sa_total_shelter_placements"

```

```

predictors <- setdiff(names(sa_reg_age), target)

formula <- as.formula(
  paste(target, "~", paste(predictors, collapse = " + "))
)

model_sa_age <- glm(formula, data = sa_reg_age, family="poisson")
summary(model_sa_age)

```

```

##
## Call:
## glm(formula = formula, family = "poisson", data = sa_reg_age)
##
## Coefficients:
##
##              Estimate Std. Error z value
## (Intercept)    6.615e+00  6.337e-02 104.385
## month2        -1.217e-02  1.764e-02  -0.690
## month3         1.121e-01  1.726e-02   6.493
## month4        -1.955e-02  1.821e-02  -1.074
## month5         2.258e-03  1.803e-02   0.125
## month6        -1.641e-02  1.813e-02  -0.905
## month7        -8.237e-02  1.753e-02  -4.698
## month8        -1.941e-02  1.737e-02  -1.118
## month9        -6.072e-02  1.764e-02  -3.443
## month10       -3.514e-02  1.739e-02  -2.020
## month11       -2.485e-02  1.757e-02  -1.415
## month12        6.449e-02  1.718e-02   3.754
## SA.Unique.Individuals.by.Age...18.thru.29  3.481e-04  2.488e-05 13.990
## SA.Unique.Individuals.by.Age...30.thru.44  -1.768e-04  2.841e-05 -6.223
## SA.Unique.Individuals.by.Age...45.thru.64   3.170e-06  8.189e-06  0.387
## SA.Unique.Individuals.by.Age...65.and.above -1.610e-04  6.300e-05 -2.556
##
##              Pr(>|z|)
## (Intercept)    < 2e-16 ***
## month2         0.490203
## month3         8.43e-11 ***
## month4         0.282958
## month5         0.900309
## month6         0.365255
## month7         2.63e-06 ***
## month8         0.263739
## month9         0.000576 ***
## month10        0.043386 *
## month11        0.157173
## month12        0.000174 ***
## SA.Unique.Individuals.by.Age...18.thru.29    < 2e-16 ***
## SA.Unique.Individuals.by.Age...30.thru.44    4.89e-10 ***
## SA.Unique.Individuals.by.Age...45.thru.64    0.698699
## SA.Unique.Individuals.by.Age...65.and.above  0.010591 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##

```

```
## Null deviance: 4421.1 on 120 degrees of freedom
## Residual deviance: 2549.4 on 105 degrees of freedom
## AIC: 3584.3
##
## Number of Fisher Scoring iterations: 4
```

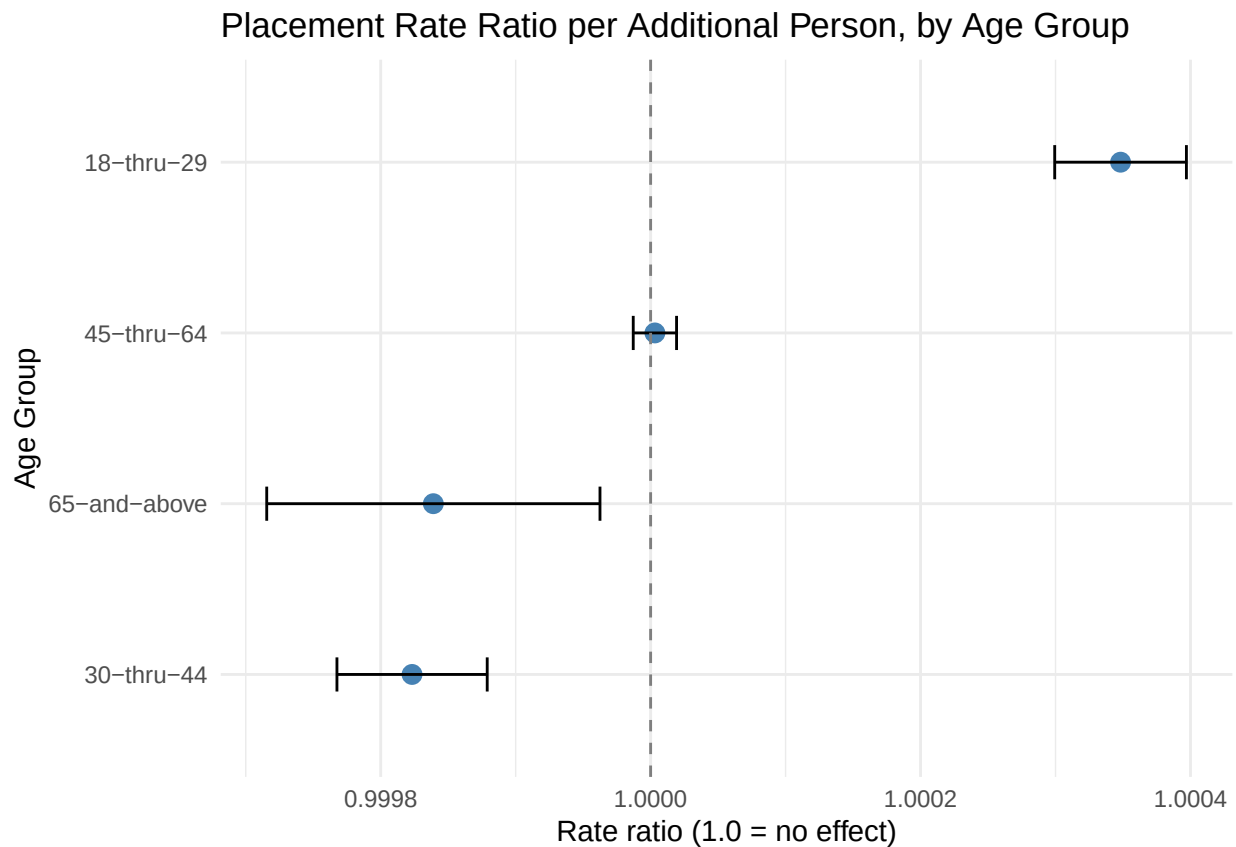
Table 5: Model fit comparison: month-only vs. month + age composition

Model	Predictors	Null Deviance	Residual Deviance	Deviance Explained	Residual df	AIC
Month only	month	5728.4	5429.1	299.3	109	6458.9
Month + Age brackets	month + 8 age brackets	5728.4	1101.2	4627.2	101	2147.1

```
library(ggplot2)
library(tidyverse)
library(broom)

age_coefs <- tidy(model_sa_age, conf.int = TRUE) %>%
  filter(grepl("^SA.Unique.Individuals.by.Age", term)) %>%
  mutate(
    term = gsub("SA.Unique.Individuals.by.Age\\\\.", "", term),
    term = gsub("\\\\.", "-", term),
    rate_ratio = exp(estimate),
    ci_low = exp(conf.low),
    ci_high = exp(conf.high)
  )

ggplot(age_coefs, aes(x = reorder(term, rate_ratio), y = rate_ratio)) +
  geom_point(size = 3, color = "steelblue") +
  geom_errorbar(aes(ymin = ci_low, ymax = ci_high), width = 0.2) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "gray50") +
  coord_flip() +
  labs(
    title = "Placement Rate Ratio per Additional Person, by Age Group",
    x = "Age Group", y = "Rate ratio (1.0 = no effect)"
  ) +
  theme_minimal()
```

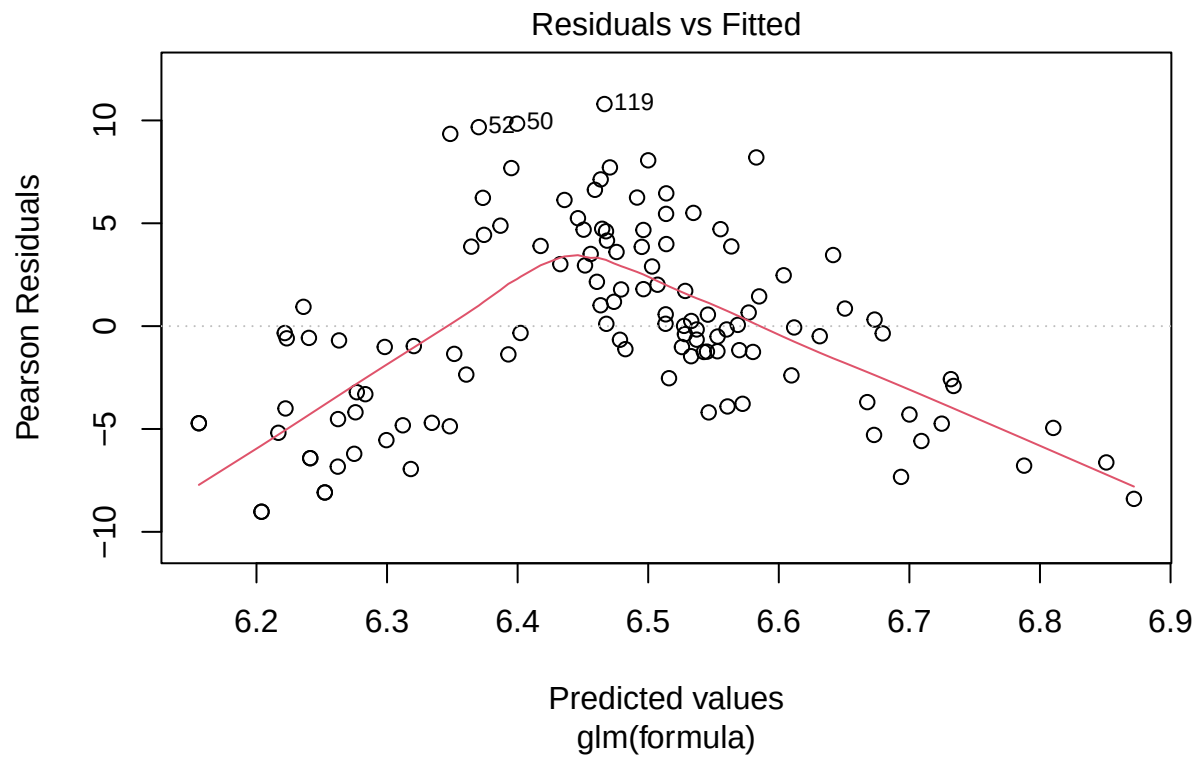


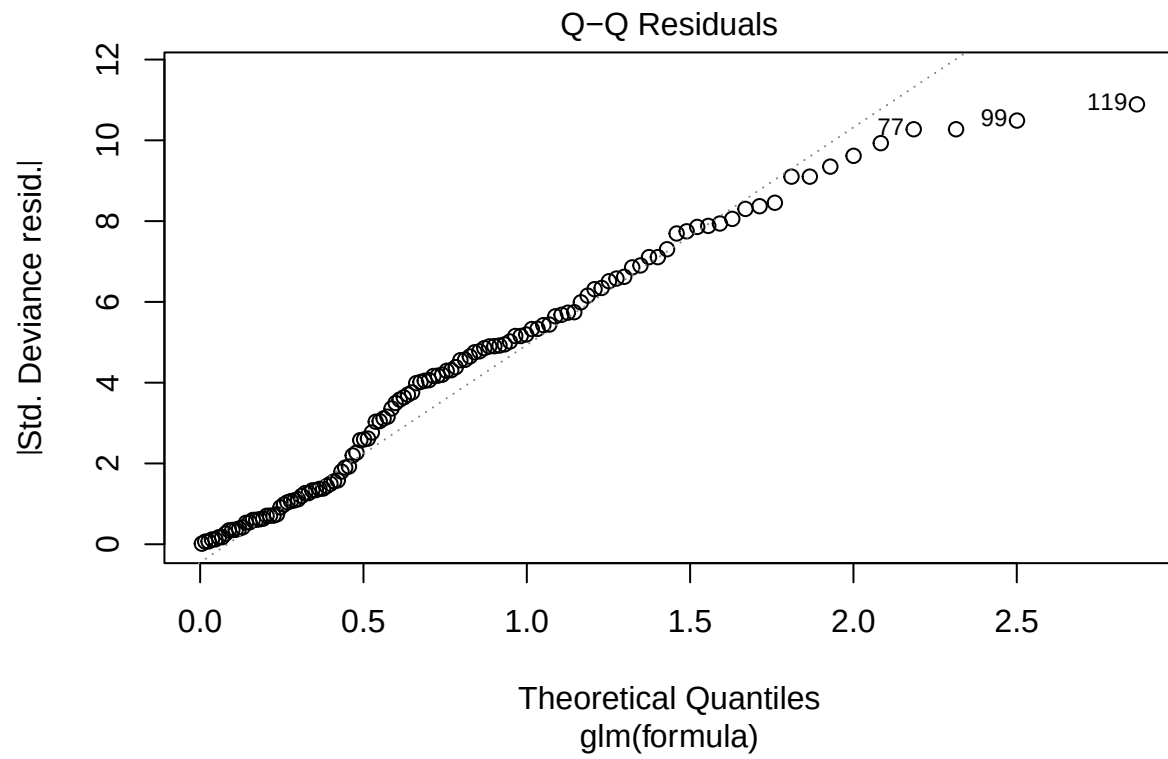
```
anova(model_sa_age, test = "Chisq")
```

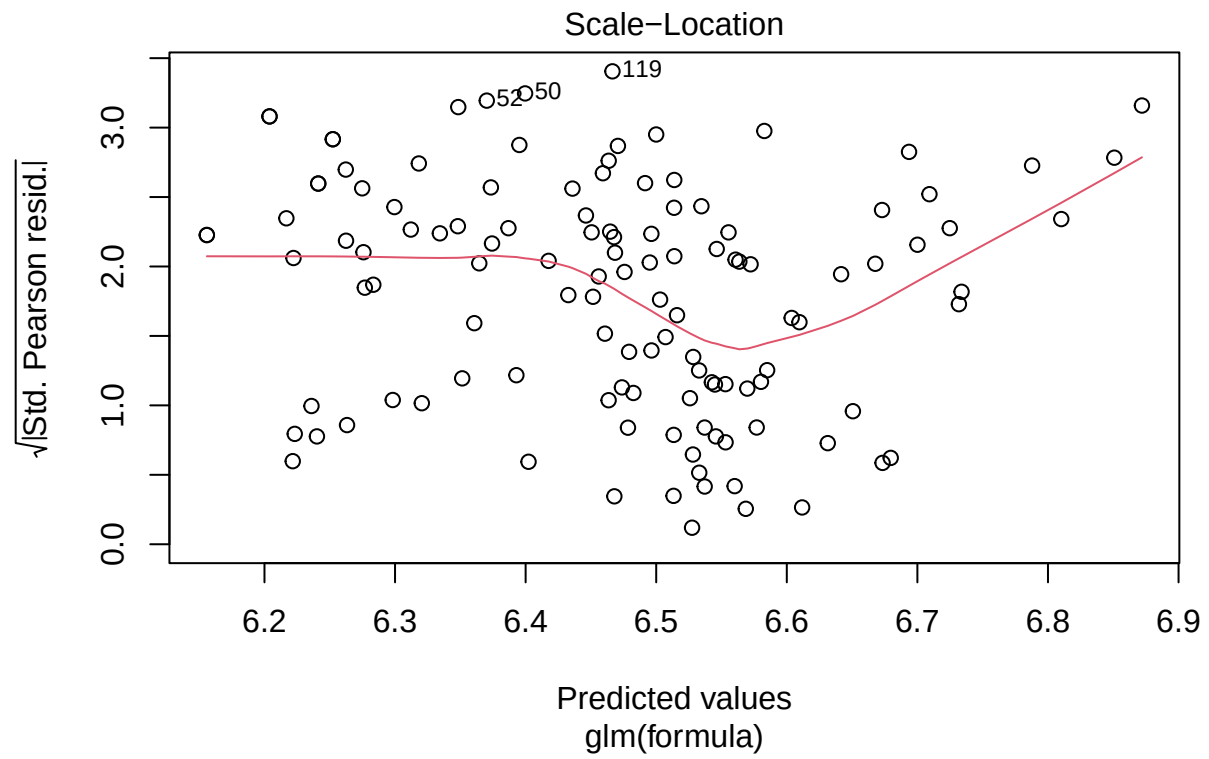
```
## Analysis of Deviance Table
##
## Model: poisson, link: log
##
## Response: sa_total_shelter_placements
##
## Terms added sequentially (first to last)
##
##
##          Df Deviance Resid. Df Resid. Dev
## NULL                                120    4421.1
## month                                11    184.98
## SA.Unique.Individuals.by.Age...18.thru.29  1    533.28
## SA.Unique.Individuals.by.Age...30.thru.44  1   1146.86
## SA.Unique.Individuals.by.Age...45.thru.64  1     0.03
## SA.Unique.Individuals.by.Age...65.and.above 1     6.53
##          Pr(>Chi)
## NULL
## month                <2e-16 ***
## SA.Unique.Individuals.by.Age...18.thru.29 <2e-16 ***
## SA.Unique.Individuals.by.Age...30.thru.44 <2e-16 ***
## SA.Unique.Individuals.by.Age...45.thru.64  0.8676
## SA.Unique.Individuals.by.Age...65.and.above 0.0106 *
```

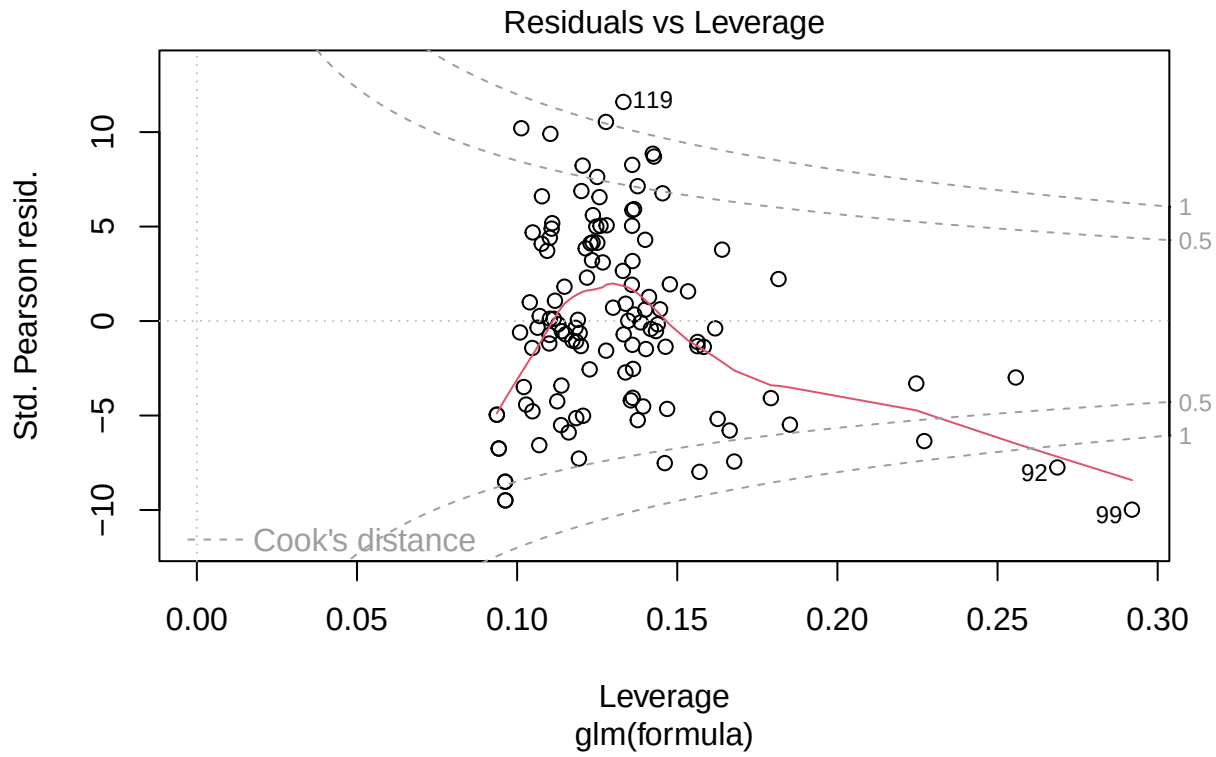
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
plot(model_sa_age)
```









```
library(car)
vif(model_sa_age)
```

##		GVIF	Df	GVIF ^{1/(2*Df)}
##	month	1.321583	11	1.012755
##	SA.Unique.Individuals.by.Age...18.thru.29	72.808769	1	8.532805
##	SA.Unique.Individuals.by.Age...30.thru.44	191.459563	1	13.836891
##	SA.Unique.Individuals.by.Age...45.thru.64	2.201643	1	1.483793
##	SA.Unique.Individuals.by.Age...65.and.above	43.354706	1	6.584429

Adult Families Ethnicity Breakdown

```
sa_reg_eth <- data_feat %>%
  select(c("month",
    "SA.Race.Ethnicity.HoH...Asian.Pacific.Islander",
    "SA.Race.Ethnicity.HoH...Black.non.Hispanic",
    "SA.Race.Ethnicity.HoH...Hispanic",
    "SA.Race.Ethnicity.HoH...Native.American",
    "SA.Race.Ethnicity.HoH...White.non.Hispanic",
    "SA.Race.Ethnicity.HoH...Unknown",
    "sa_total_shelter_placements") %>%
    na.omit())

target <- "sa_total_shelter_placements"
```

```

predictors <- setdiff(names(sa_reg_eth), target)

formula <- as.formula(
  paste(target, "~", paste(predictors, collapse = " + "))
)

model_sa_eth <- glm(formula, data = sa_reg_eth, family="poisson")
summary(model_sa_eth)

```

```

##
## Call:
## glm(formula = formula, family = "poisson", data = sa_reg_eth)
##
## Coefficients:
##
##              Estimate Std. Error z value
## (Intercept)    6.275e+00  3.601e-02 174.249
## month2        -1.977e-02  1.768e-02  -1.118
## month3         9.357e-02  1.726e-02   5.421
## month4         2.016e-02  1.821e-02   1.107
## month5         6.569e-02  1.813e-02   3.623
## month6         5.396e-03  1.811e-02   0.298
## month7        -4.450e-02  1.764e-02  -2.522
## month8         1.150e-02  1.726e-02   0.666
## month9        -1.091e-02  1.774e-02  -0.615
## month10        2.990e-02  1.735e-02   1.724
## month11        1.079e-02  1.750e-02   0.616
## month12        8.312e-02  1.764e-02   4.713
## SA.Race.Ethnicity.HoH...Asian.Pacific.Islander  3.740e-03  1.874e-04  19.960
## SA.Race.Ethnicity.HoH...Black.non.Hispanic    -5.958e-05  5.382e-06 -11.071
## SA.Race.Ethnicity.HoH...Hispanic             -8.793e-06  3.562e-06  -2.468
## SA.Race.Ethnicity.HoH...Native.American      -2.674e-03  2.045e-03  -1.308
## SA.Race.Ethnicity.HoH...White.non.Hispanic    2.843e-04  4.033e-05   7.050
## SA.Race.Ethnicity.HoH...Unknown              -4.221e-05  2.396e-05  -1.762
##
##              Pr(>|z|)
## (Intercept)    < 2e-16 ***
## month2         0.263632
## month3         5.93e-08 ***
## month4         0.268168
## month5         0.000291 ***
## month6         0.765766
## month7         0.011669 *
## month8         0.505268
## month9         0.538523
## month10        0.084726 .
## month11        0.537722
## month12        2.44e-06 ***
## SA.Race.Ethnicity.HoH...Asian.Pacific.Islander  < 2e-16 ***
## SA.Race.Ethnicity.HoH...Black.non.Hispanic    < 2e-16 ***
## SA.Race.Ethnicity.HoH...Hispanic             0.013575 *
## SA.Race.Ethnicity.HoH...Native.American      0.190882
## SA.Race.Ethnicity.HoH...White.non.Hispanic    1.79e-12 ***
## SA.Race.Ethnicity.HoH...Unknown              0.078148 .
## ---

```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##    Null deviance: 4419.2  on 119  degrees of freedom
## Residual deviance: 3340.2  on 102  degrees of freedom
##    (1 observation deleted due to missingness)
## AIC: 4370.7
##
## Number of Fisher Scoring iterations: 4
```

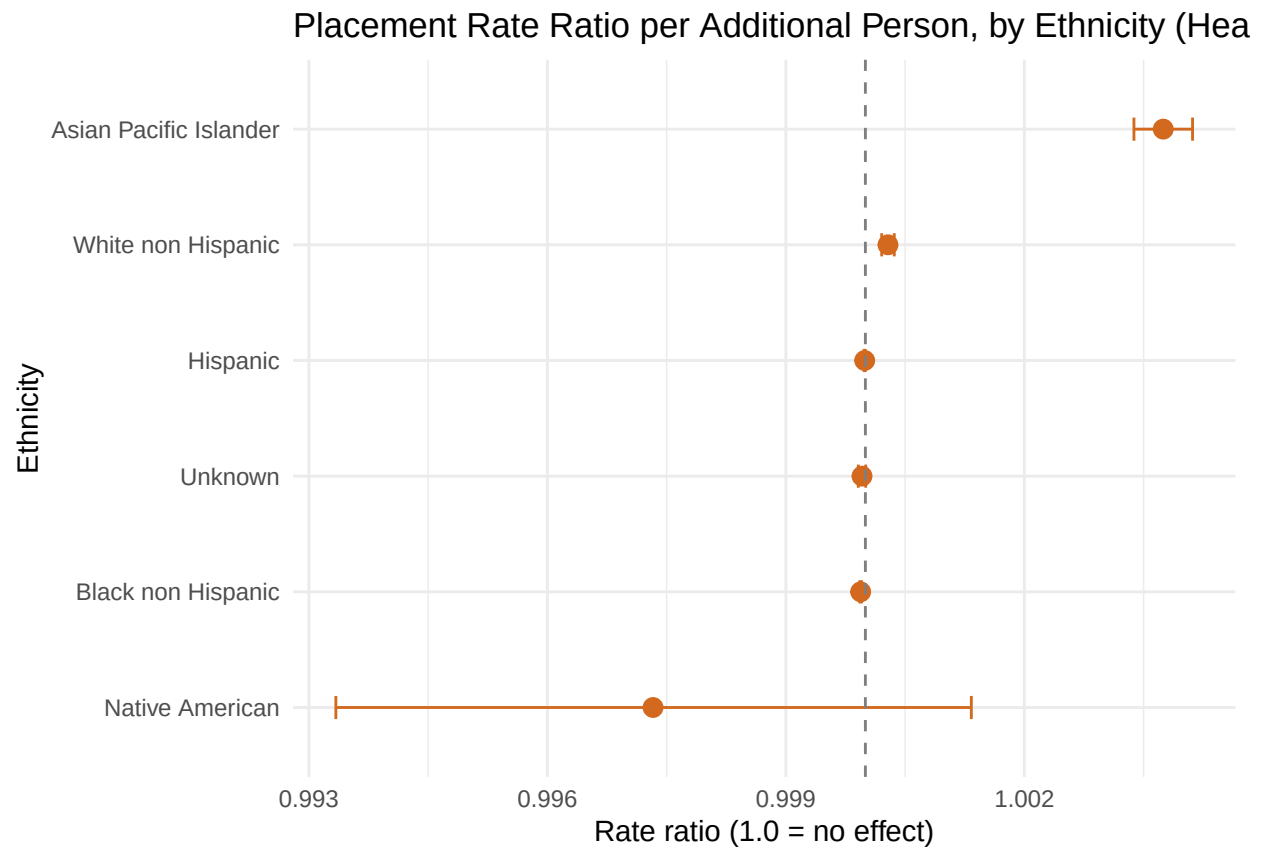
Table 6: Model fit comparison: month-only vs. month + ethnicity composition

Model	Predictors	Null Deviance	Residual Deviance	Deviance Explained	Residual df	AIC
Month only	month	5728.4	5429.1	299.3	109	6458.9
Month + Age brackets	month + 8 age brackets	5728.4	1101.3	4627.1	103	2053.2

```
library(ggplot2)
library(broom)
library(dplyr)

eth_coefs <- tidy(model_sa_eth, conf.int = TRUE) %>%
  filter(grepl("^SA.Race.Ethnicity.HoH", term)) %>%
  mutate(
    term = gsub("SA.Race.Ethnicity.HoH\\.", "", term),
    term = gsub("\\.", " ", term),
    rate_ratio = exp(estimate),
    ci_low = exp(conf.low),
    ci_high = exp(conf.high)
  )

ggplot(eth_coefs, aes(x = reorder(term, rate_ratio), y = rate_ratio)) +
  geom_point(size = 3, color = "#D2691E") +
  geom_errorbar(aes(ymin = ci_low, ymax = ci_high), width = 0.2, color = "#D2691E") +
  geom_hline(yintercept = 1, linetype = "dashed", color = "gray50") +
  coord_flip() +
  labs(
    title = "Placement Rate Ratio per Additional Person, by Ethnicity (Head of Household)",
    x = "Ethnicity", y = "Rate ratio (1.0 = no effect)"
  ) +
  theme_minimal()
```



Single Adult Breakdown by Gender

```
sa_reg_gdr <- data_feat %>%
  select(c("month",
           "SA.Avg.Daily.census.Individual.in.Shelter...Men",
           "SA.Avg.Daily.census.Individual.in.Shelter...Women",
           "af_total_shelter_placements") %>%
  na.omit())

target <- "af_total_shelter_placements"

predictors <- setdiff(names(sa_reg_gdr), target)

formula <- as.formula(
  paste(target, "~", paste(predictors, collapse = " + "))
)

model_sa_gdr <- glm(formula, data = sa_reg_gdr, family="poisson")
summary(model_sa_gdr)
```

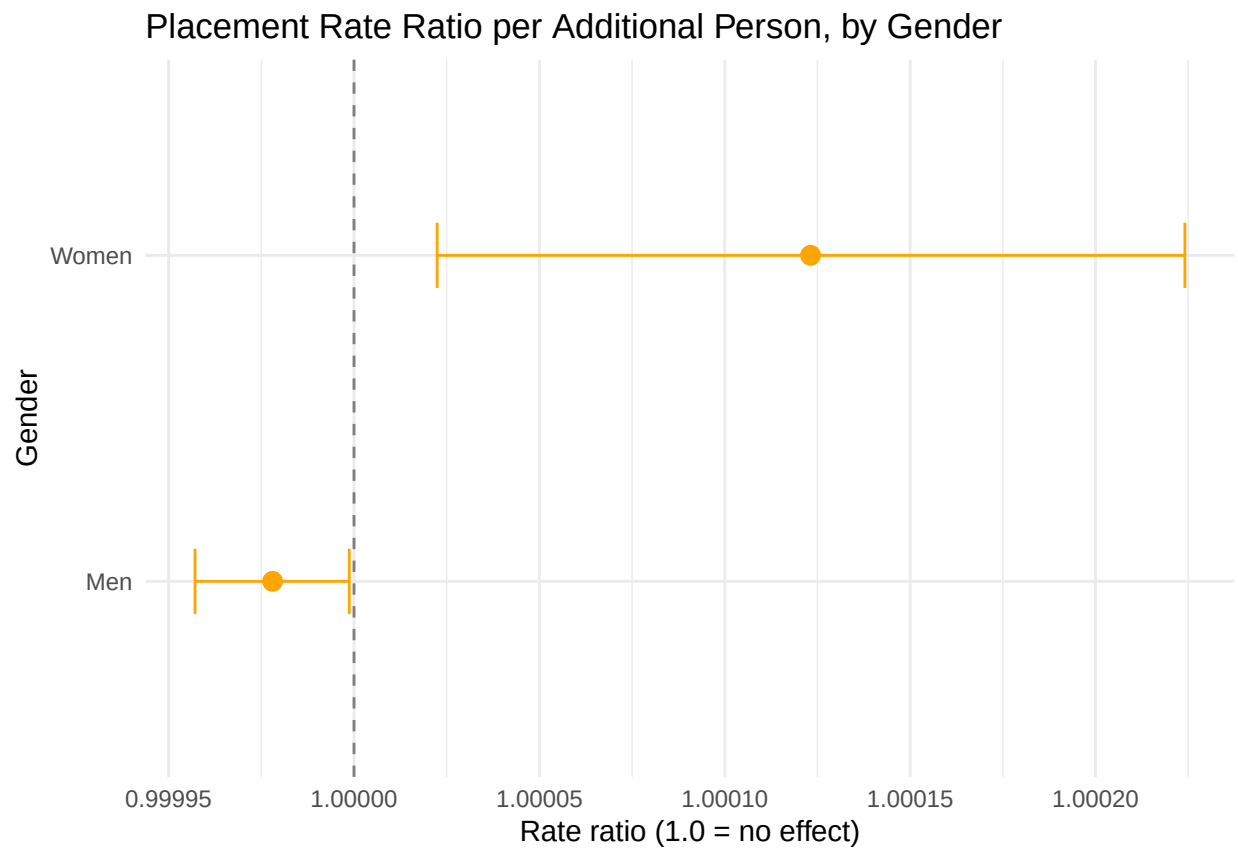
```
##
## Call:
## glm(formula = formula, family = "poisson", data = sa_reg_gdr)
##
## Coefficients:
##                                     Estimate Std. Error z value
```

```
## (Intercept) 3.608e+00 1.426e-01 25.302
## month2 -3.262e-02 6.507e-02 -0.501
## month3 1.818e-03 6.452e-02 0.028
## month4 -1.189e-01 6.891e-02 -1.725
## month5 6.467e-02 6.578e-02 0.983
## month6 2.578e-02 6.631e-02 0.389
## month7 3.294e-02 6.317e-02 0.521
## month8 -3.921e-02 6.410e-02 -0.612
## month9 -3.762e-02 6.385e-02 -0.589
## month10 -1.379e-01 6.530e-02 -2.112
## month11 -1.182e-01 6.651e-02 -1.777
## month12 -9.681e-03 6.469e-02 -0.150
## SA.Avg.Daily.census.Individual.in.Shelter...Men -2.195e-05 1.061e-05 -2.069
## SA.Avg.Daily.census.Individual.in.Shelter...Women 1.231e-04 5.144e-05 2.394
## Pr(>|z|)
## (Intercept) <2e-16 ***
## month2 0.6161
## month3 0.9775
## month4 0.0845 .
## month5 0.3256
## month6 0.6974
## month7 0.6021
## month8 0.5408
## month9 0.5557
## month10 0.0347 *
## month11 0.0756 .
## month12 0.8810
## SA.Avg.Daily.census.Individual.in.Shelter...Men 0.0385 *
## SA.Avg.Daily.census.Individual.in.Shelter...Women 0.0167 *
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 441.85 on 120 degrees of freedom
## Residual deviance: 417.15 on 107 degrees of freedom
## AIC: 1127.4
##
## Number of Fisher Scoring iterations: 4
```

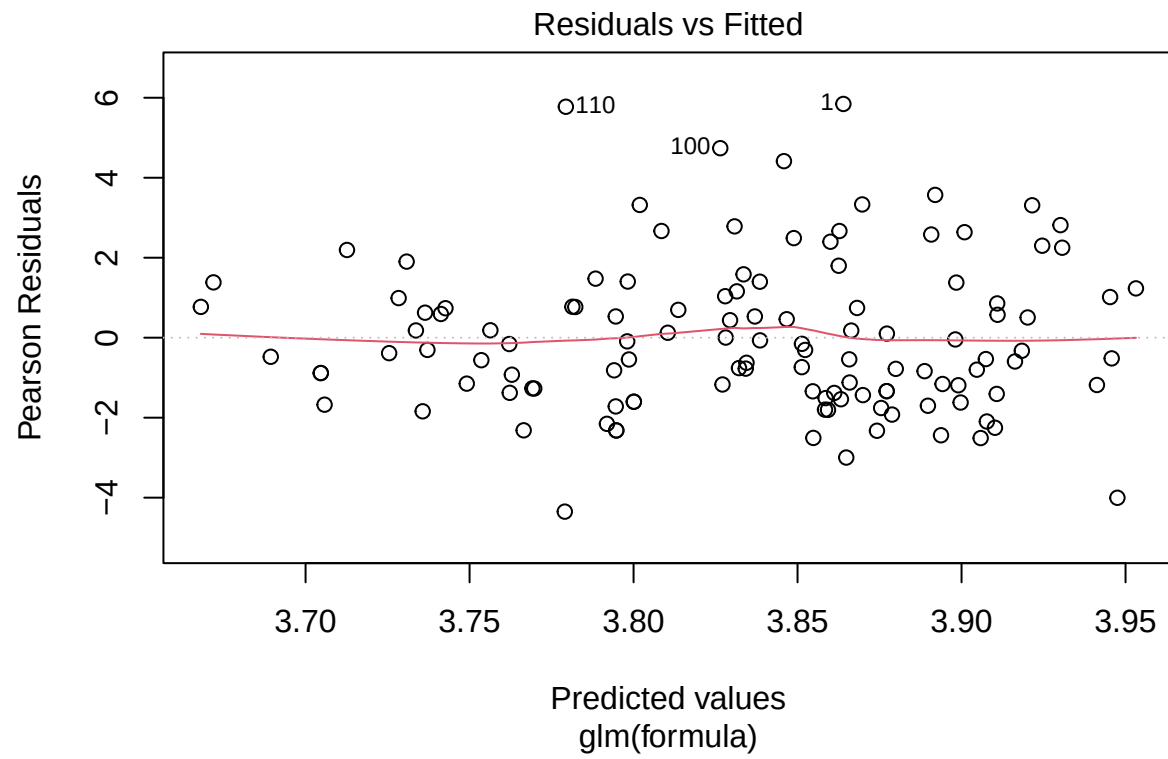
```
library(ggplot2)
library(broom)
library(dplyr)

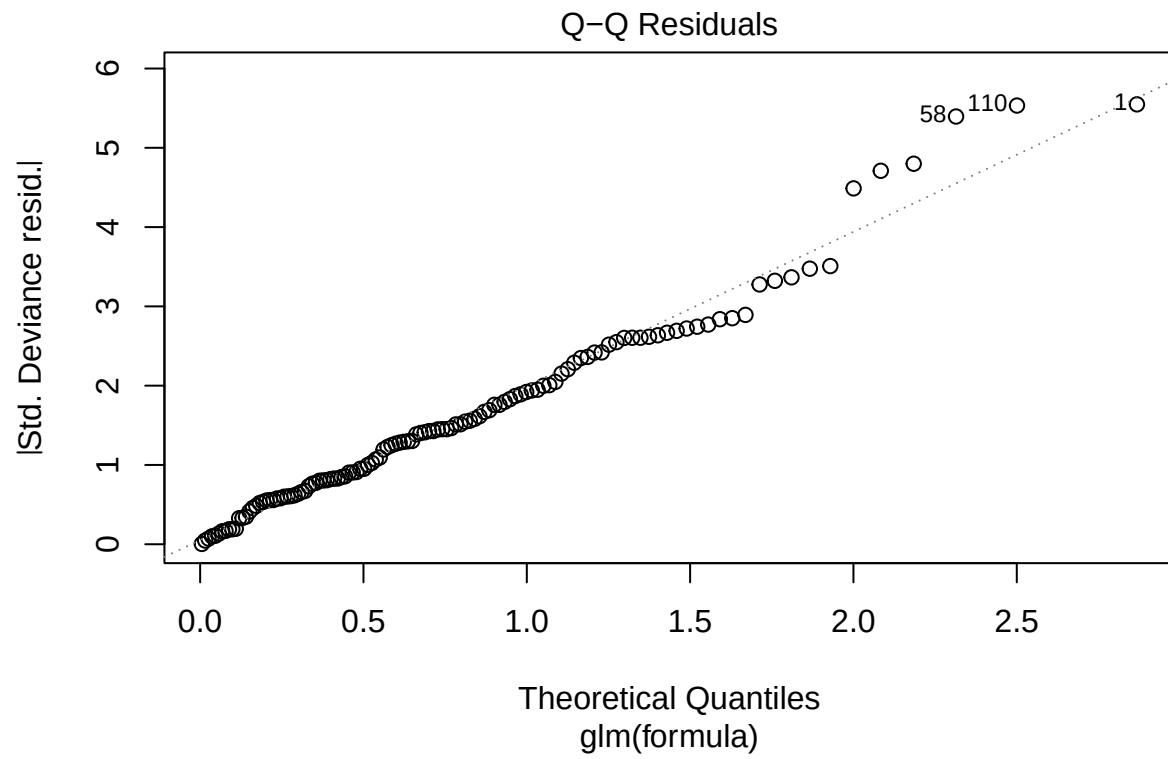
gender_coefs <- tidy(model_sa_gdr, conf.int = TRUE) %>%
  filter(grepl("^SA.Avg.Daily.census.Individual.in.Shelter", term)) %>%
  mutate(
    term = gsub("SA.Avg.Daily.census.Individual.in.Shelter\\.", "", term),
    term = gsub("\\.", " ", term),
    rate_ratio = exp(estimate),
    ci_low = exp(conf.low),
    ci_high = exp(conf.high)
  )
```

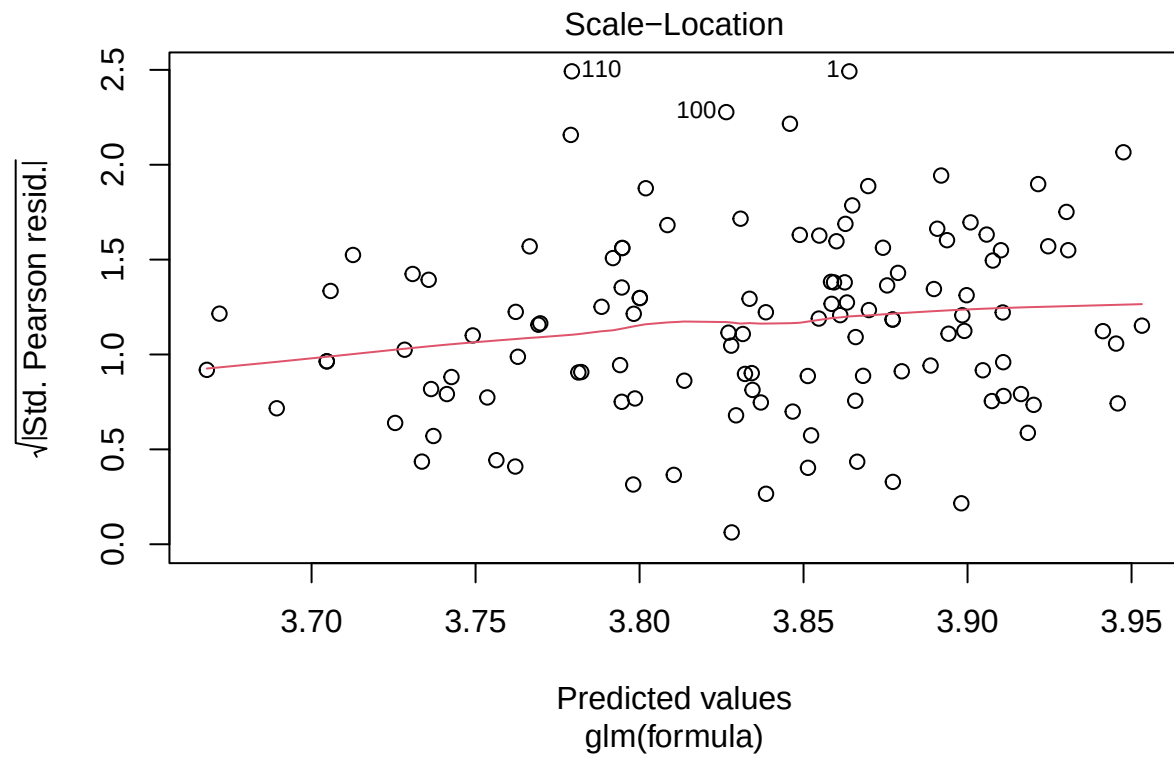
```
ggplot(gender_coefs, aes(x = reorder(term, rate_ratio), y = rate_ratio)) +
  geom_point(size = 3, color = "orange") +
  geom_errorbar(aes(ymin = ci_low, ymax = ci_high), width = 0.2, color = "orange") +
  geom_hline(yintercept = 1, linetype = "dashed", color = "gray50") +
  coord_flip() +
  labs(
    title = "Placement Rate Ratio per Additional Person, by Gender",
    x = "Gender", y = "Rate ratio (1.0 = no effect)"
  ) +
  theme_minimal()
```

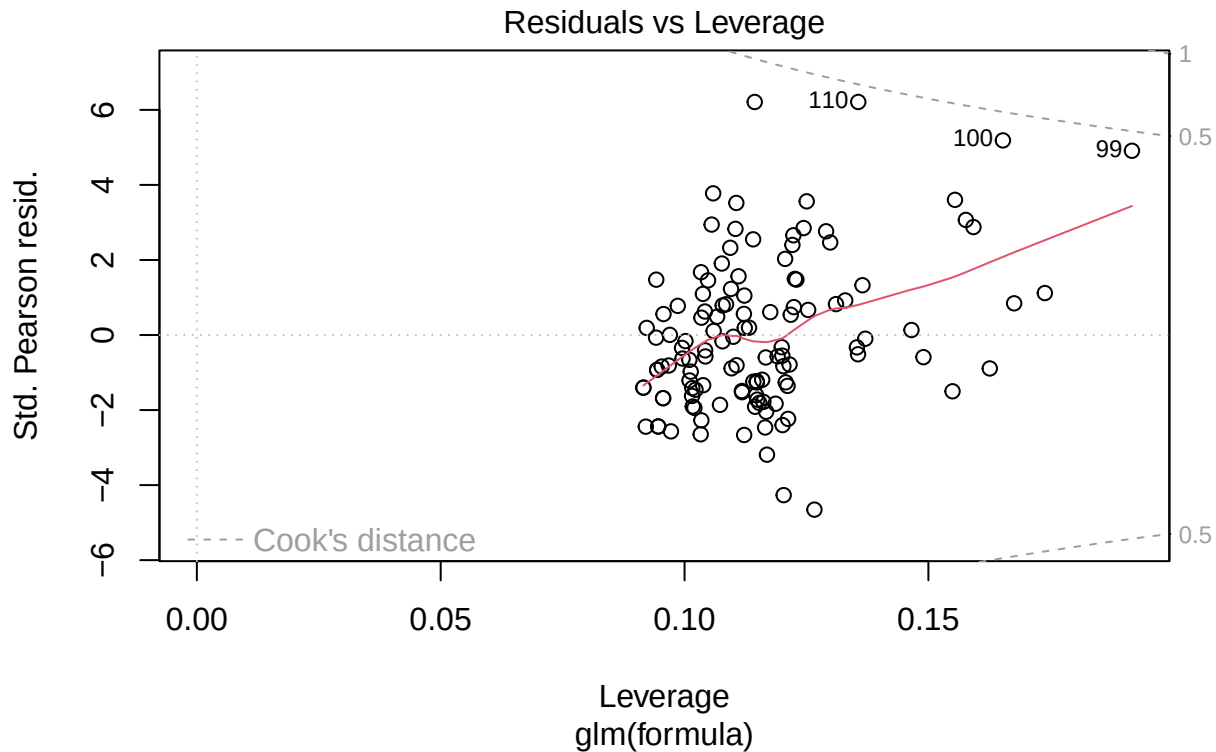


```
plot(model_sa_gdr)
```









```
## =====
## FWC age regression: raw counts vs. proportions (unmerged)
## vs. proportions (30-44 + 45-64 merged)
## Proportions expressed in PERCENTAGE POINTS (0-100), not
## raw proportions (0-1), so coefficients are directly readable
## as "rate ratio per 1 percentage point increase" -- no manual
## rescaling needed after the fact.
## =====

library(dplyr)
library(car)

## -----
## 0. Assumes `data_feat` already exists, same as before.
## -----

age_cols <- c(
  "FWC.Unique.Individuals.by.Age...0.thru.5",
  "FWC.Unique.Individuals.by.Age...6.thru.13",
  "FWC.Unique.Individuals.by.Age...14.thru.17",
  "FWC.Unique.Individuals.by.Age...18.thru.20",
  "FWC.Unique.Individuals.by.Age...21.thru.29",
  "FWC.Unique.Individuals.by.Age...30.thru.44",
  "FWC.Unique.Individuals.by.Age...45.thru.64",
  "FWC.Unique.Individuals.by.Age...65.and.above"
)
```

```

## -----
## 1. MODEL 1: raw counts, no offset (original, kept for reference)
## -----

fwc_reg_age_raw <- data_feat %>%
  select(all_of(c("month", age_cols, "fwc_total_shelter_placements"))) %>%
  na.omit()

model_raw <- glm(
  fwc_total_shelter_placements ~ .,
  data = fwc_reg_age_raw,
  family = "poisson"
)

## -----
## 2. MODEL 2: proportions (percentage points), all brackets,
##    21-29 dropped as reference, offset for total population
## -----

fwc_reg_age_prop <- data_feat %>%
  select(all_of(c("month", age_cols, "fwc_total_shelter_placements"))) %>%
  na.omit() %>%
  mutate(fwc_total_population = rowSums(across(all_of(age_cols)))) %>%
  mutate(across(all_of(age_cols), ~ 100 * .x / fwc_total_population, .names = "pctpt_{.col}")) %>%
  select(month, starts_with("pctpt_"), fwc_total_population, fwc_total_shelter_placements) %>%
  select(-`pctpt_FWC.Unique.Individuals.by.Age...21.thru.29`)

model_prop <- glm(
  fwc_total_shelter_placements ~ . - fwc_total_population
  + offset(log(fwc_total_population)),
  data = fwc_reg_age_prop,
  family = "poisson"
)

## -----
## 3. MODEL 3: proportions (percentage points), 30-44 and 45-64
##    merged -- kept for comparison, but note this fit WORSE
##    than model 2 on deviance/AIC in earlier testing, so treat
##    model 2 as the primary candidate unless this changes that.
## -----

fwc_reg_age_merged <- data_feat %>%
  select(all_of(c("month", age_cols, "fwc_total_shelter_placements"))) %>%
  na.omit() %>%
  mutate(
    fwc_total_population = rowSums(across(all_of(age_cols))),
    adults_30_64 = `FWC.Unique.Individuals.by.Age...30.thru.44` +
      `FWC.Unique.Individuals.by.Age...45.thru.64`
  ) %>%
  mutate(
    pctpt_0_5   = 100 * `FWC.Unique.Individuals.by.Age...0.thru.5` / fwc_total_population,
    pctpt_6_13  = 100 * `FWC.Unique.Individuals.by.Age...6.thru.13` / fwc_total_population,
    pctpt_14_17 = 100 * `FWC.Unique.Individuals.by.Age...14.thru.17` / fwc_total_population,

```

```

pctpt_18_20 = 100 * `FWC.Unique.Individuals.by.Age..18.thru.20` / fwc_total_population,
pctpt_30_64 = 100 * adults_30_64 / fwc_total_population,
pctpt_65up = 100 * `FWC.Unique.Individuals.by.Age...65.and.above` / fwc_total_population
# 21-29 dropped again as the reference category
) %>%
select(month, starts_with("pctpt_"), fwc_total_population, fwc_total_shelter_placements)

model_merged <- glm(
  fwc_total_shelter_placements ~ . - fwc_total_population
  + offset(log(fwc_total_population)),
  data = fwc_reg_age_merged,
  family = "poisson"
)

## -----
## 4. VIF comparison
## -----

cat("\n===== VIF: raw model =====\n")

```

```

##
## ===== VIF: raw model =====

```

```
print(vif(model_raw))
```

```

##
##              GVIF Df  GVIF^(1/(2*Df))
## month              1.608392 11      1.021837
## FWC.Unique.Individuals.by.Age...0.thru.5    433.074554 1      20.810443
## FWC.Unique.Individuals.by.Age...6.thru.13    222.243876 1      14.907846
## FWC.Unique.Individuals.by.Age...14.thru.17    466.639558 1      21.601842
## FWC..Unique.Individuals.by.Age..18.thru.20    145.286727 1      12.053494
## FWC.Unique.Individuals.by.Age...21.thru.29    104.042749 1      10.200135
## FWC.Unique.Individuals.by.Age...30.thru.44     85.663538 1       9.255460
## FWC.Unique.Individuals.by.Age...45.thru.64    123.806709 1      11.126846
## FWC.Unique.Individuals.by.Age...65.and.above   14.408088 1       3.795799

```

```
cat("\n===== VIF: proportions, unmerged (pct points) =====\n")
```

```

##
## ===== VIF: proportions, unmerged (pct points) =====

```

```
print(vif(model_prop))
```

```

##
##              GVIF Df  GVIF^(1/(2*Df))
## month              1.297901 11      1.011923
## pctpt_FWC.Unique.Individuals.by.Age...0.thru.5    17.776572 1       4.216227
## pctpt_FWC.Unique.Individuals.by.Age...6.thru.13     8.203462 1       2.864169
## pctpt_FWC.Unique.Individuals.by.Age...14.thru.17     9.355100 1       3.058611
## pctpt_FWC..Unique.Individuals.by.Age..18.thru.20     2.902796 1       1.703759
## pctpt_FWC.Unique.Individuals.by.Age...30.thru.44    36.962436 1       6.079674
## pctpt_FWC.Unique.Individuals.by.Age...45.thru.64    14.033030 1       3.746069
## pctpt_FWC.Unique.Individuals.by.Age...65.and.above   3.202991 1       1.789690

```

```
cat("\n===== VIF: proportions, merged (pct points) =====\n")
```

```
##
```

```
## ===== VIF: proportions, merged (pct points) =====
```

```
print(vif(model_merged))
```

```
##              GVIF Df GVIF^(1/(2*Df))
## month          1.236668 11          1.009702
## pctpt_0_5      18.086279 1          4.252797
## pctpt_6_13     8.164752 1          2.857403
## pctpt_14_17    3.898734 1          1.974521
## pctpt_18_20    2.351410 1          1.533431
## pctpt_30_64   23.832345 1          4.881838
## pctpt_65up     3.263180 1          1.806427
```

```
cat("\n===== Max VIF summary =====\n")
```

```
##
```

```
## ===== Max VIF summary =====
```

```
cat("Raw model max VIF:           ", max(vif(model_raw)[, 1]), "\n")
```

```
## Raw model max VIF:           466.6396
```

```
cat("Proportions (unmerged) max VIF:", max(vif(model_prop)[, 1]), "\n")
```

```
## Proportions (unmerged) max VIF: 36.96244
```

```
cat("Proportions (merged) max VIF:  ", max(vif(model_merged)[, 1]), "\n")
```

```
## Proportions (merged) max VIF:  23.83234
```

```
## -----
## 5. Fit comparison: deviance, AIC, dispersion
## -----
```

```
cat("\n===== Deviance / dispersion comparison =====\n")
```

```
##
```

```
## ===== Deviance / dispersion comparison =====
```

```
comparison <- data.frame(
  Model = c("Raw counts", "Proportions (unmerged)", "Proportions (merged)"),
  Null_Deviance = c(model_raw$null.deviance, model_prop$null.deviance, model_merged$null.deviance),
  Residual_Deviance = c(deviance(model_raw), deviance(model_prop), deviance(model_merged)),
  Resid_df = c(df.residual(model_raw), df.residual(model_prop), df.residual(model_merged)),
  AIC = c(AIC(model_raw), AIC(model_prop), AIC(model_merged)),
```

```

Dispersion_ratio = c(
  deviance(model_raw) / df.residual(model_raw),
  deviance(model_prop) / df.residual(model_prop),
  deviance(model_merged) / df.residual(model_merged)
)
)
print(comparison)

```

```

##           Model Null_Deviance Residual_Deviance Resid_df      AIC
## 1           Raw counts      5728.426          1101.217     101 2147.096
## 2 Proportions (unmerged)      3958.348          1156.089     102 2199.968
## 3 Proportions (merged)      3958.348          1450.158     103 2492.037
## Dispersion_ratio
## 1           10.90314
## 2           11.33420
## 3           14.07921

```

```

cat("\nNote: dispersion ratio well above 1 for all three models means\n")

```

```

##
## Note: dispersion ratio well above 1 for all three models means

```

```

cat("standard errors and p-values are likely too optimistic regardless\n")

```

```

## standard errors and p-values are likely too optimistic regardless

```

```

cat("of which model you pick -- worth addressing with quasipoisson or\n")

```

```

## of which model you pick -- worth addressing with quasipoisson or

```

```

cat("negative binomial as a next step.\n")

```

```

## negative binomial as a next step.

```

```

## -----
## 6. Coefficients as rate ratios PER 1 PERCENTAGE POINT
##    (directly interpretable now -- no rescaling needed)
## -----

```

```

cat("\n===== Proportions (unmerged) coefficients =====\n")

```

```

##
## ===== Proportions (unmerged) coefficients =====

```

```

prop_coefs <- data.frame(
  term = names(coef(model_prop)),
  estimate = coef(model_prop),
  rate_ratio_per_1pp = exp(coef(model_prop))
)
print(prop_coefs, row.names = FALSE)

```

```
##               term      estimate
##      (Intercept) -17.95089926
##               month2  -0.14047097
##               month3  -0.04985488
##               month4  -0.05531426
##               month5  -0.02609410
##               month6  -0.01789719
##               month7  -0.10914275
##               month8  -0.05370694
##               month9  -0.11328533
##               month10 -0.11503332
##               month11 -0.18735795
##               month12 -0.03126236
##      pctpt_FWC.Unique.Individuals.by.Age...0.thru.5  0.21199625
##      pctpt_FWC.Unique.Individuals.by.Age...6.thru.13  0.08681282
##      pctpt_FWC.Unique.Individuals.by.Age...14.thru.17 -0.06352174
##      pctpt_FWC..Unique.Individuals.by.Age...18.thru.20  0.17324138
##      pctpt_FWC.Unique.Individuals.by.Age...30.thru.44  0.16881549
##      pctpt_FWC.Unique.Individuals.by.Age...45.thru.64  0.65078755
##      pctpt_FWC.Unique.Individuals.by.Age...65.and.above  1.08093864
##      rate_ratio_per_1pp
##           1.599645e-08
##           8.689489e-01
##           9.513675e-01
##           9.461878e-01
##           9.742434e-01
##           9.822620e-01
##           8.966024e-01
##           9.477098e-01
##           8.928959e-01
##           8.913364e-01
##           8.291469e-01
##           9.692213e-01
##           1.236143e+00
##           1.090693e+00
##           9.384537e-01
##           1.189153e+00
##           1.183902e+00
##           1.917050e+00
##           2.947445e+00
```

```
cat("\n===== Proportions (merged) coefficients =====\n")
```

```
##
## ===== Proportions (merged) coefficients =====
```

```
merged_coefs <- data.frame(
  term = names(coef(model_merged)),
  estimate = coef(model_merged),
  rate_ratio_per_1pp = exp(coef(model_merged))
)
print(merged_coefs, row.names = FALSE)
```

##	term	estimate	rate_ratio_per_1pp
##	(Intercept)	-1.657955e+01	6.303667e-08
##	month2	-1.307849e-01	8.774065e-01
##	month3	-2.551317e-02	9.748095e-01
##	month4	-3.249899e-02	9.680234e-01
##	month5	-1.844838e-02	9.817208e-01
##	month6	-4.718692e-04	9.995282e-01
##	month7	-7.607755e-02	9.267443e-01
##	month8	-5.303523e-02	9.483466e-01
##	month9	-1.138958e-01	8.923509e-01
##	month10	-1.128139e-01	8.933169e-01
##	month11	-1.904147e-01	8.266163e-01
##	month12	-4.445598e-02	9.565177e-01
##	pctpt_0_5	2.183172e-01	1.243982e+00
##	pctpt_6_13	8.910255e-02	1.093193e+00
##	pctpt_14_17	3.884711e-01	1.474724e+00
##	pctpt_18_20	-2.119017e-01	8.090442e-01
##	pctpt_30_64	1.056187e-01	1.111398e+00
##	pctpt_65up	8.477764e-01	2.334450e+00